Toward World Models: Geometry, View Synthesis, and Visual Reasoning

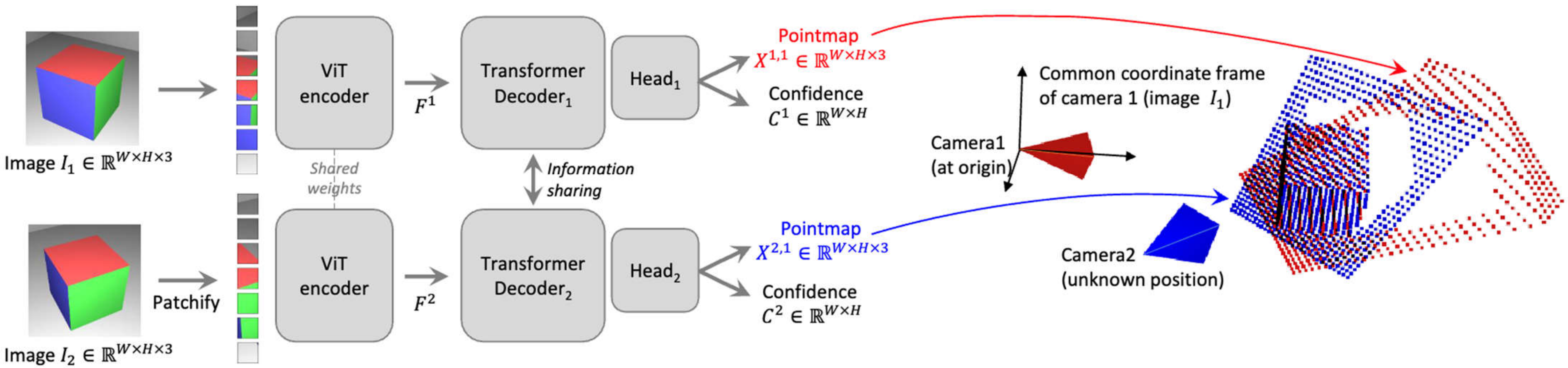
Ming-Hsuan Yang

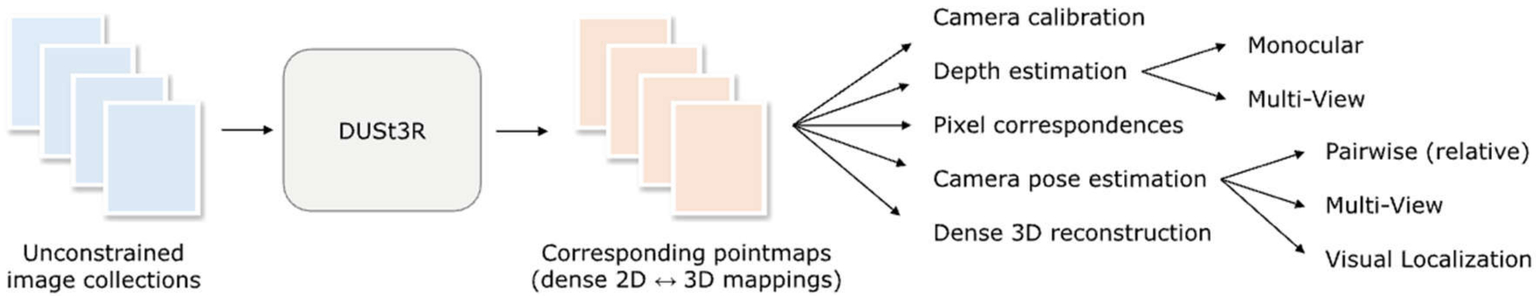
UC Merced Google DeepMind

Perceive and Generate 3D Scenes

- Advances in novel view synthesis, Gaussian splats, depth estimation, large-reconstruction models, diffusion models, …
- Exploit relationships between 2D images and 3D world (e.g., point map) from one to multiple views
- Formulate numerous vision tasks in one model based on
  - Monocular projective geometry
  - Multiview geometry
  - Depth estimation
  - Camera motion

Multiview Geometry with World Coordinate





All about depth in one world coordinate

Dynamic 4D Scenes

- World is dynamic
- Research trend
  - Static scenes -> Deformable dynamic scenes
  - Posed images -> Arbitrary images
  - Optimization method -> Feedforward model
  - Real-time performance
  - High-quality, high-fidelity representation

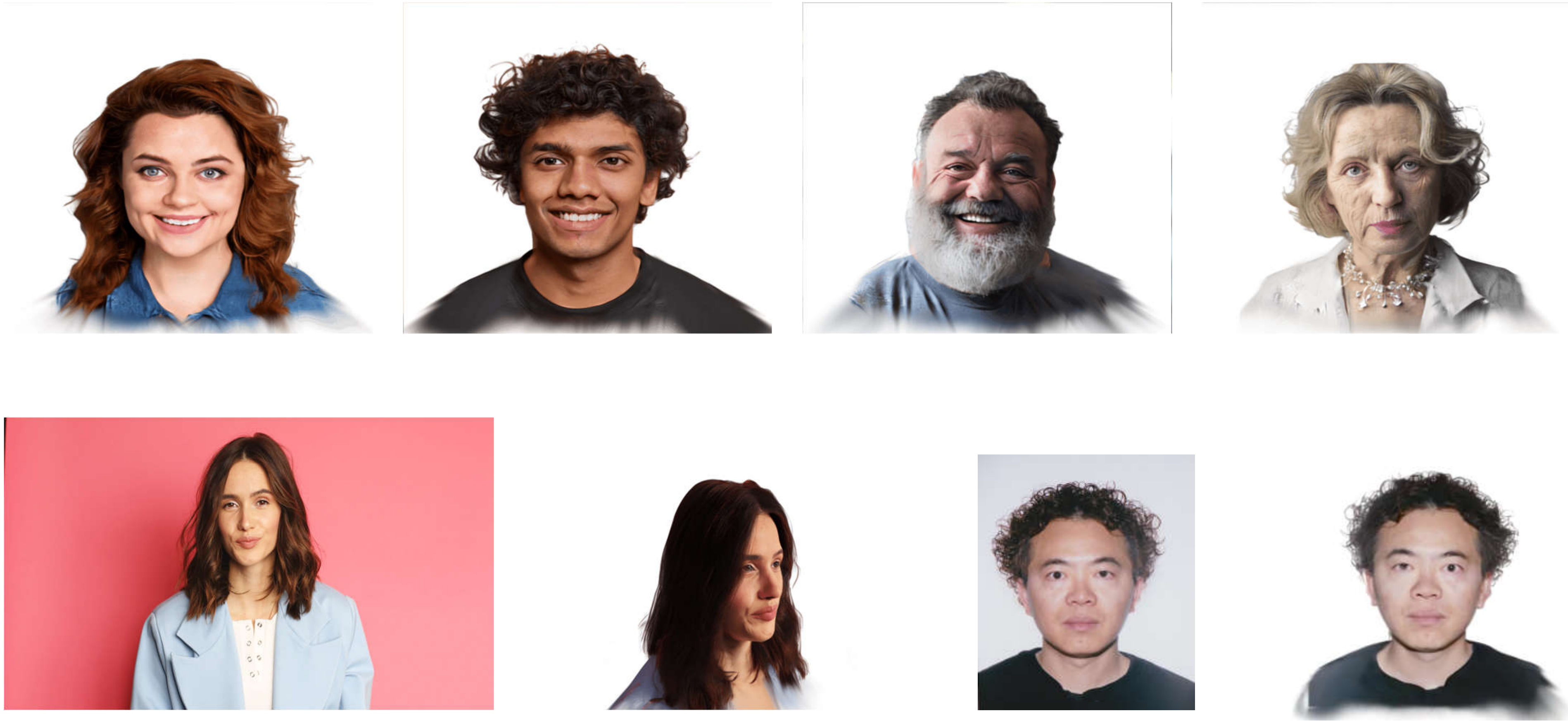
MonST3R [ICLR 2025]

Handling dynamic objects (prior to VGGT and $\pi 3$)

NoPoSplat [ICLR 2025]

NoPoSplat: Gaussian splatting without pose or significant overlap

FaceLift [ICCV 2025]



Two stage framework based on GS-LRM and diffusion model

From Creating Views to Understanding Worlds

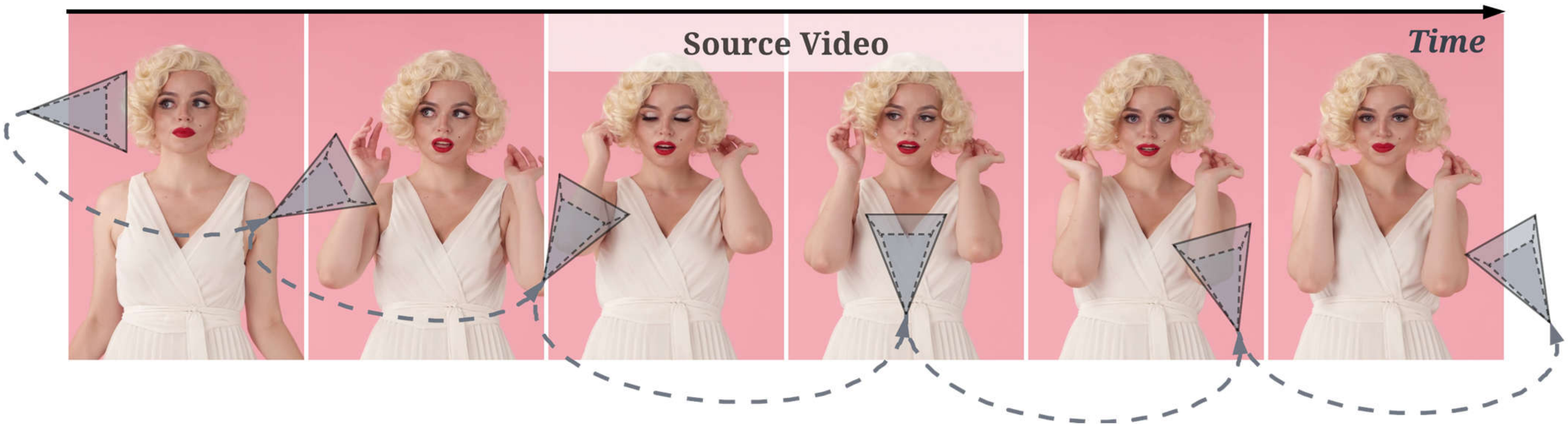
- Generation and understanding

  What I cannot create, I do not understand --- Richard Feynman

- A world model who *understands* can:
  - Control camera views
  - Interactively control motions
  - Remember and generate long context
  - Accept and reconstruct with streaming input

Topics

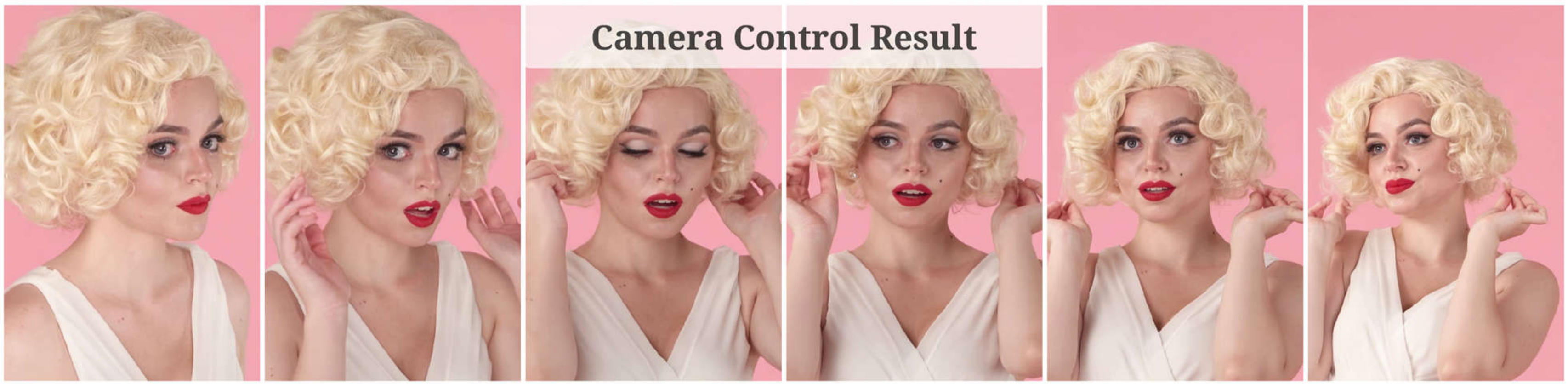
- FaceCam [CVPR 2026]
- MotiMotion [ICML 2026]
- Context Forcing [ICML 2026]
- ReCoSplat [In Submission]



FaceCam: Portrait Video Camera Control via Scale-Aware Conditioning

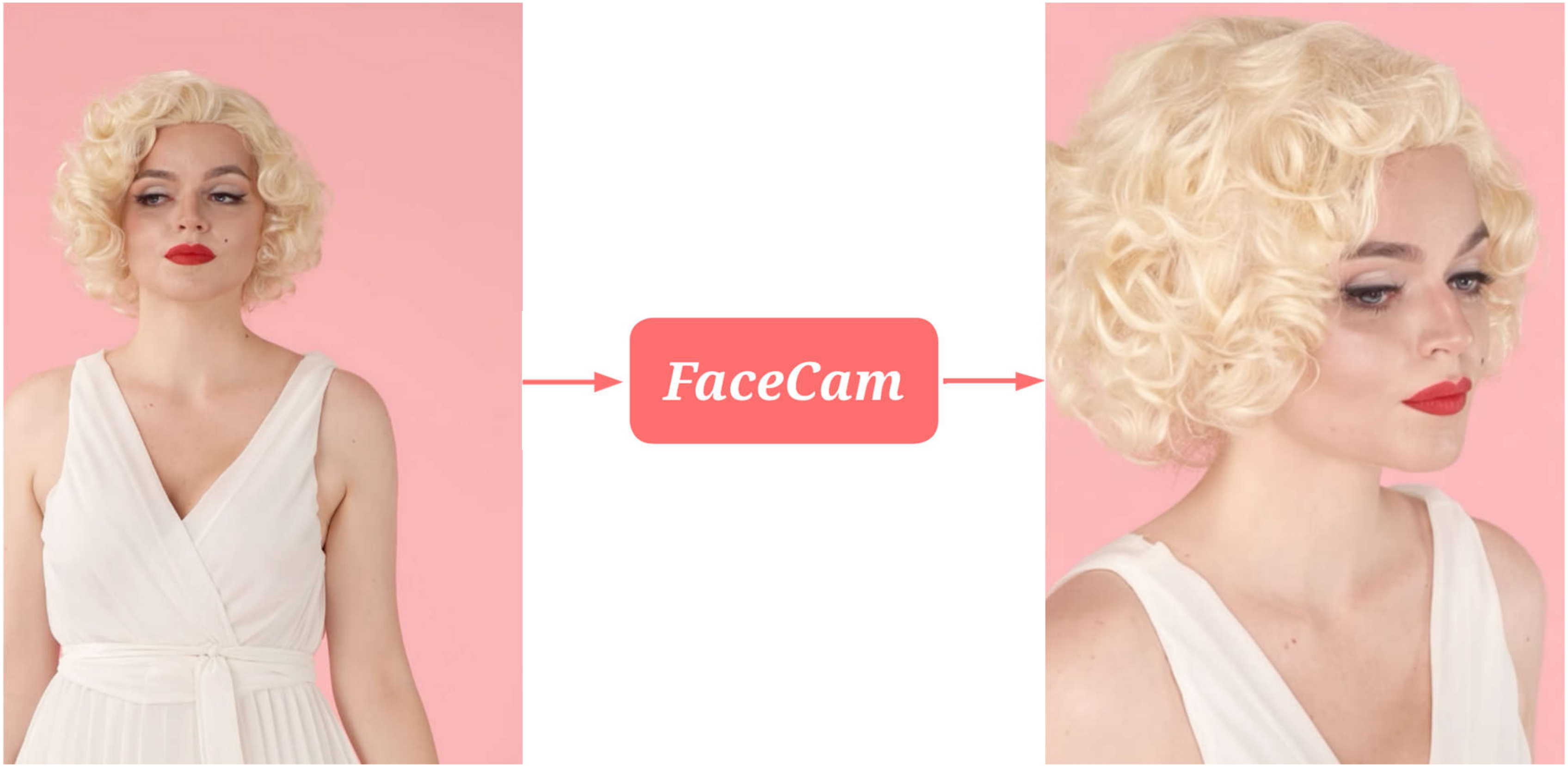
CVPR 2026

Weijie Lyu Ming-Hsuan Yang Zhixin Shu



University of California, Merced Adobe Research

Overview



Source Video

Camera Control Result

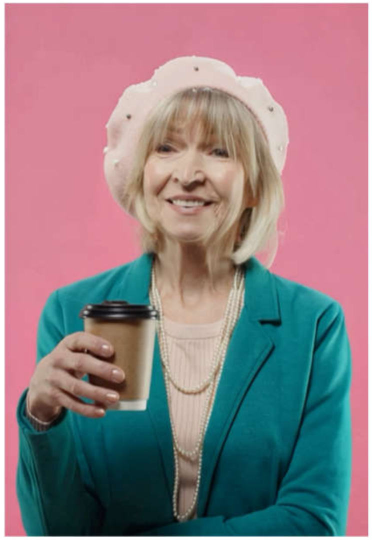
Existing Methods

Reconstruction-based camera representation

- *Condition on a rendered dynamic point cloud*
- *Reconstruction errors propagate into facial distortion and identify drift*

Source Video

Point Cloud Rendering



Generated Video

FaceCam

YU, Mark, et al. "Trajectorycrafter: Redirecting camera trajectory for monocular videos via diffusion models." *In ICCV 2025.*

Existing Methods

Parameter-based camera representation

- *Condition on camera extrinsic parameters*
- *Scale-ambiguous problem causes inaccurate camera control*

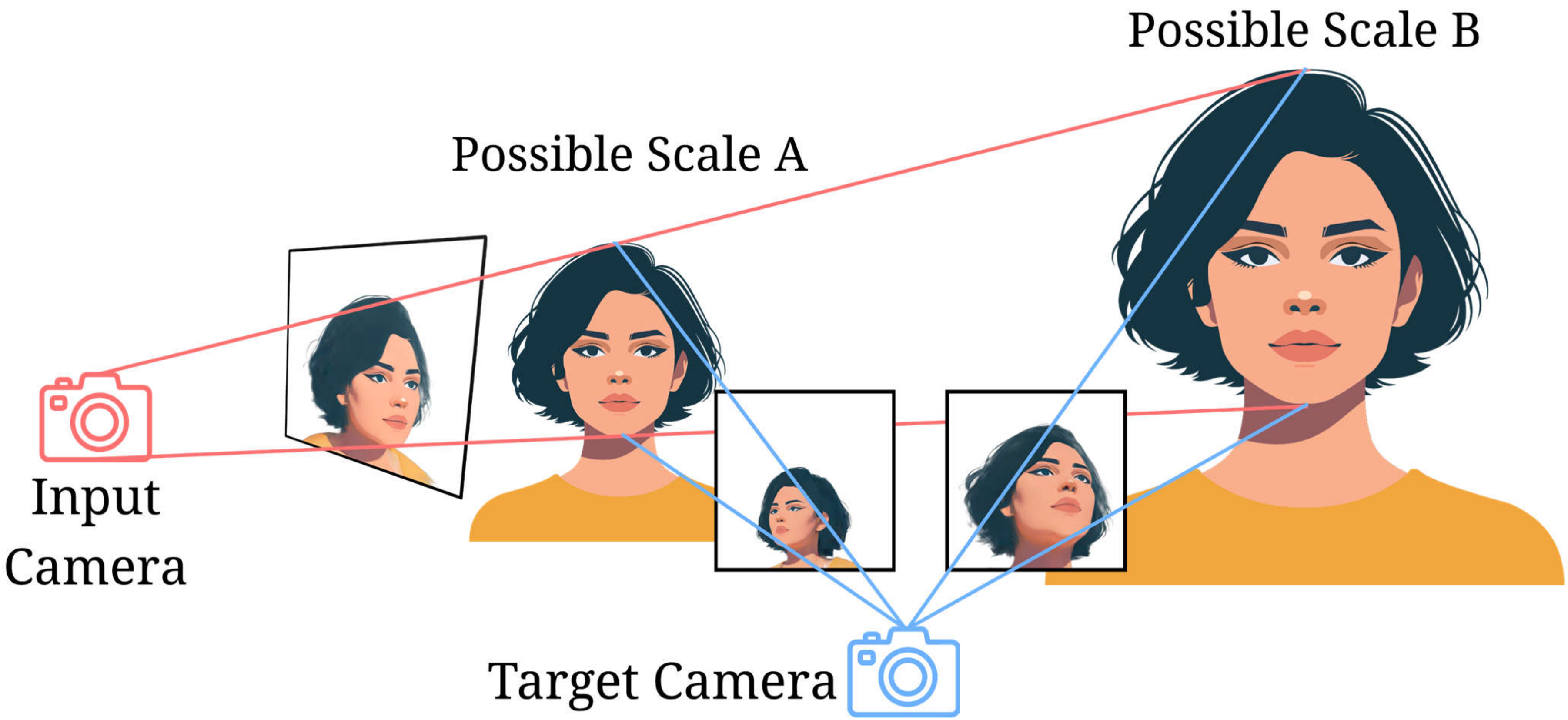
Rotate 30 degrees *Rotate 30 degrees*

Source Video | Generated Video | *FaceCam* | Source Video | Generated Video | *FaceCam*

Bai, Jianhong, et al. "Recammaster: Camera-controlled generative rendering from a single video." *In ICCV 2025*.

Existing Methods

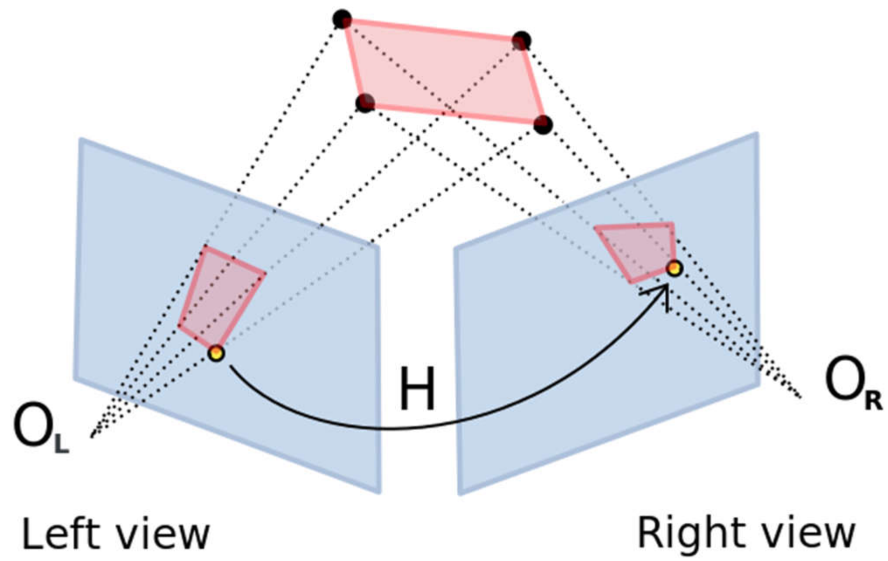
Camera parameters and monocular scale ambiguity



Scale-Aware Camera Conditioning

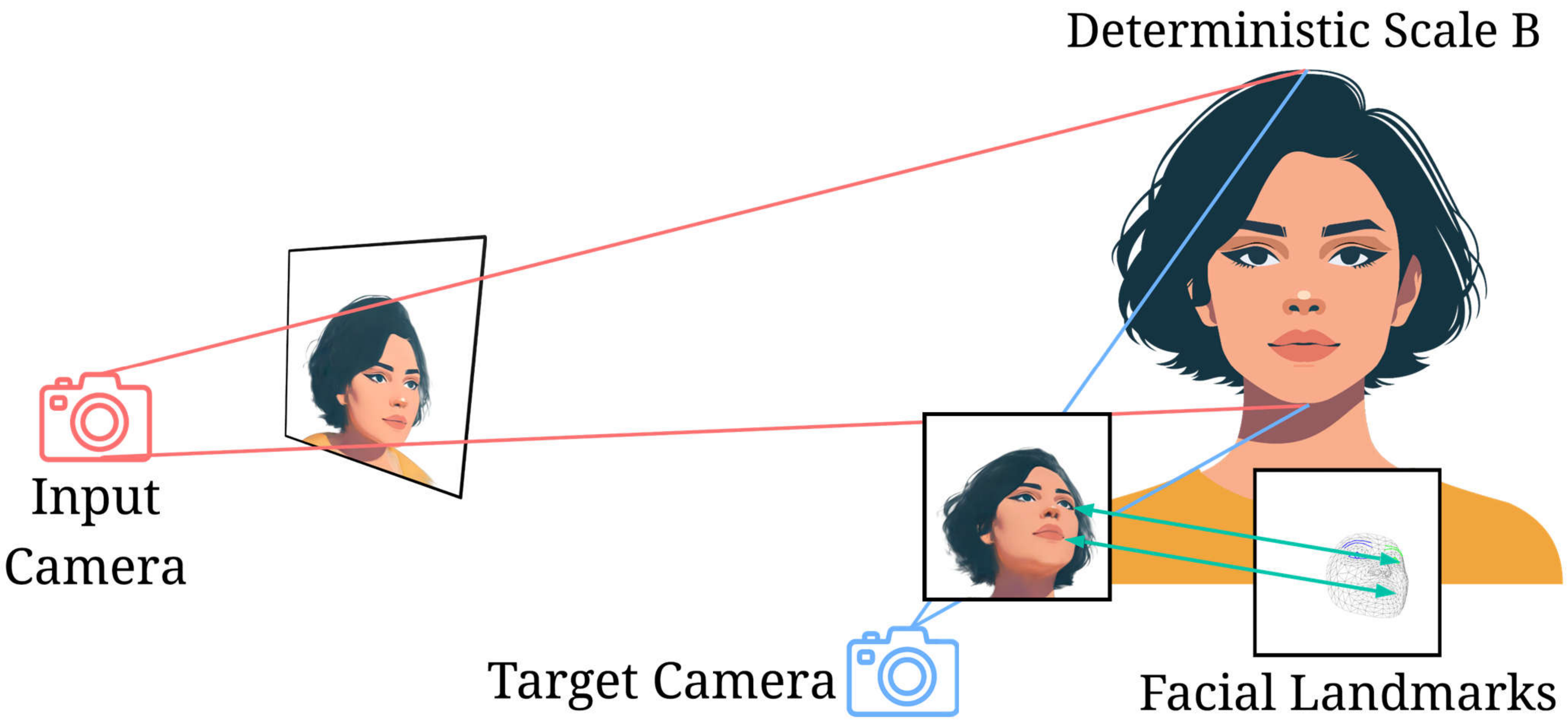
Camera conditioning using image-space point correspondences

- *Given two views and a set of corresponding pixels, estimate a fundamental matrix that satisfies the epipolar constraint*



Scale-Aware Camera Conditioning

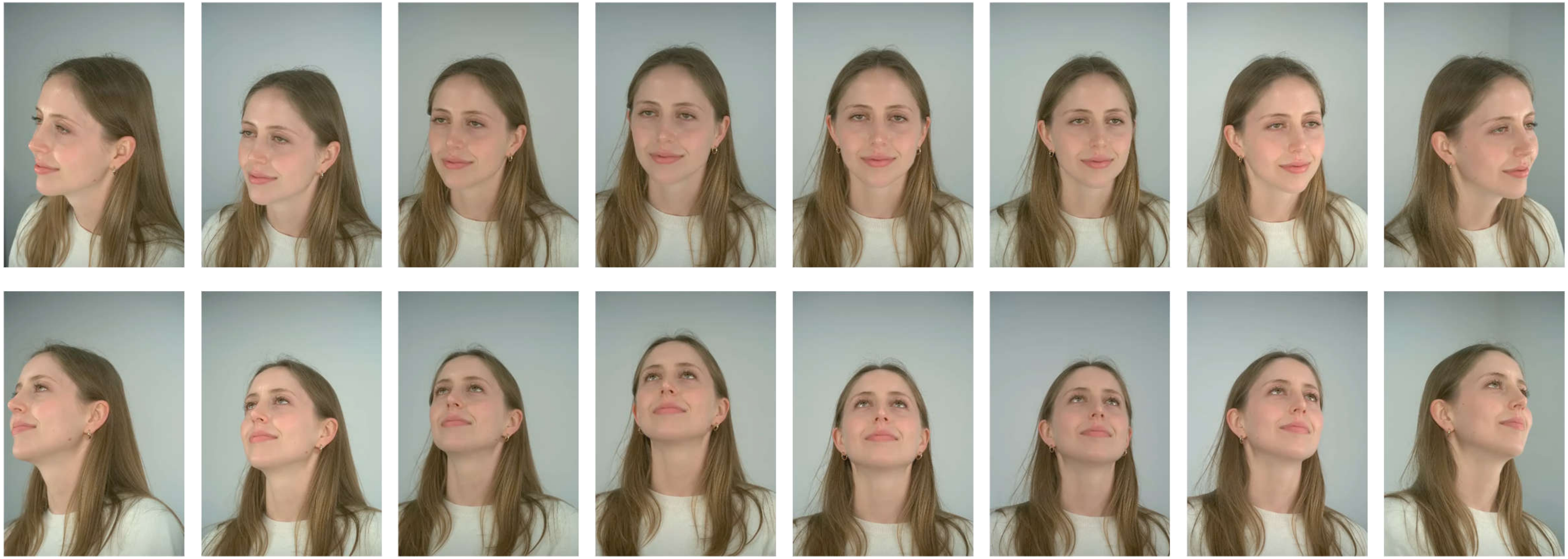
Camera conditioning using facial landmarks



Training Data

Studio-captured multi-view dataset

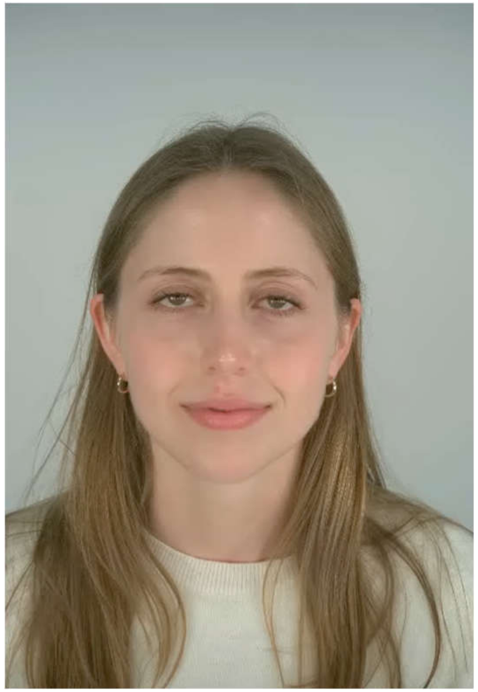
425 subjects, 16 viewpoints, 9.4K video sequences in total



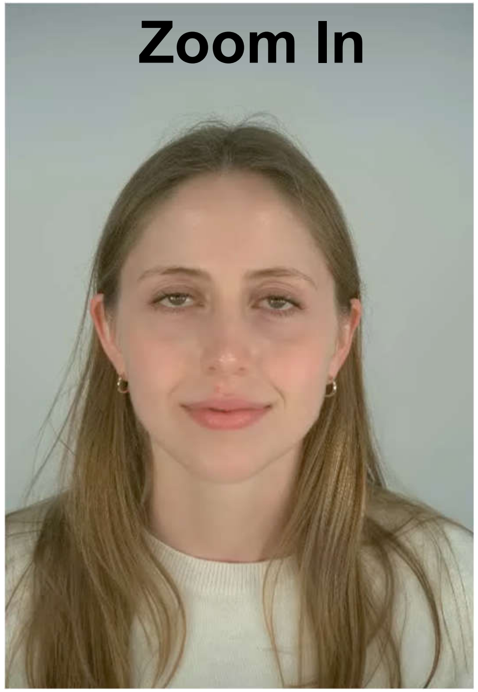
Data Augumentation

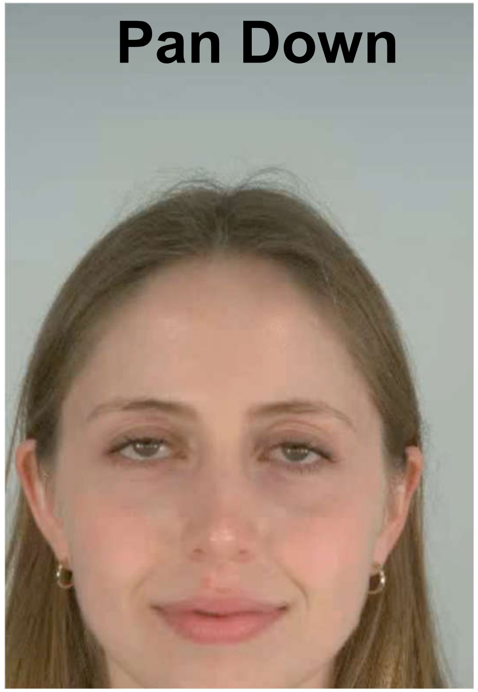
Adding camera control

Original Video



Synthetic Camera Motion

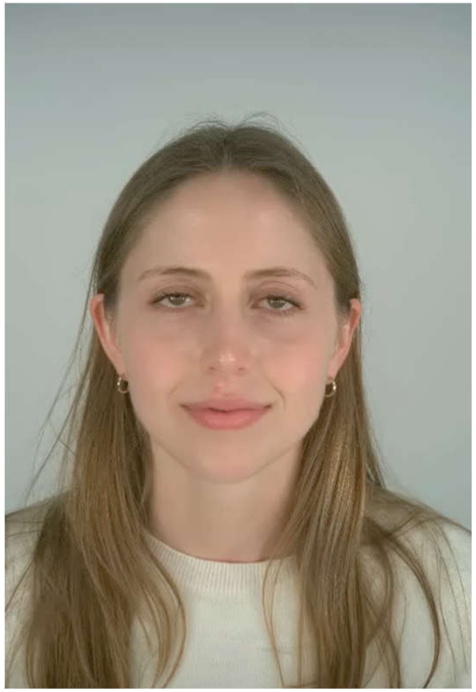




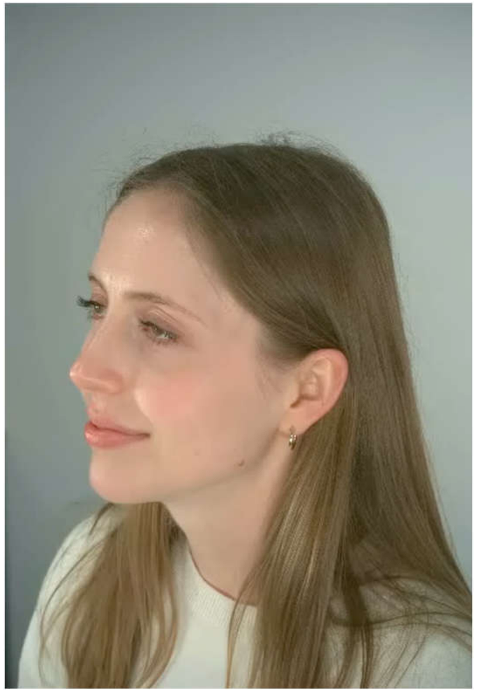
Data Augumentation

Adding camera control

Original Video



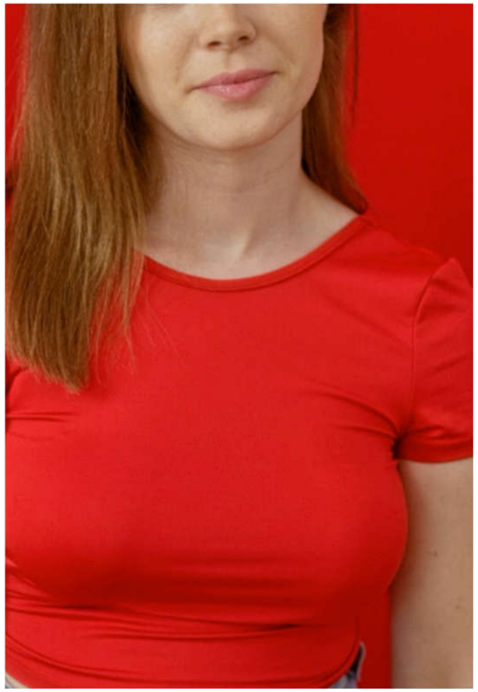
Multi-shot Stitching



Data Augumentation

In-the-wild videos w. Synthetic Camera Motion

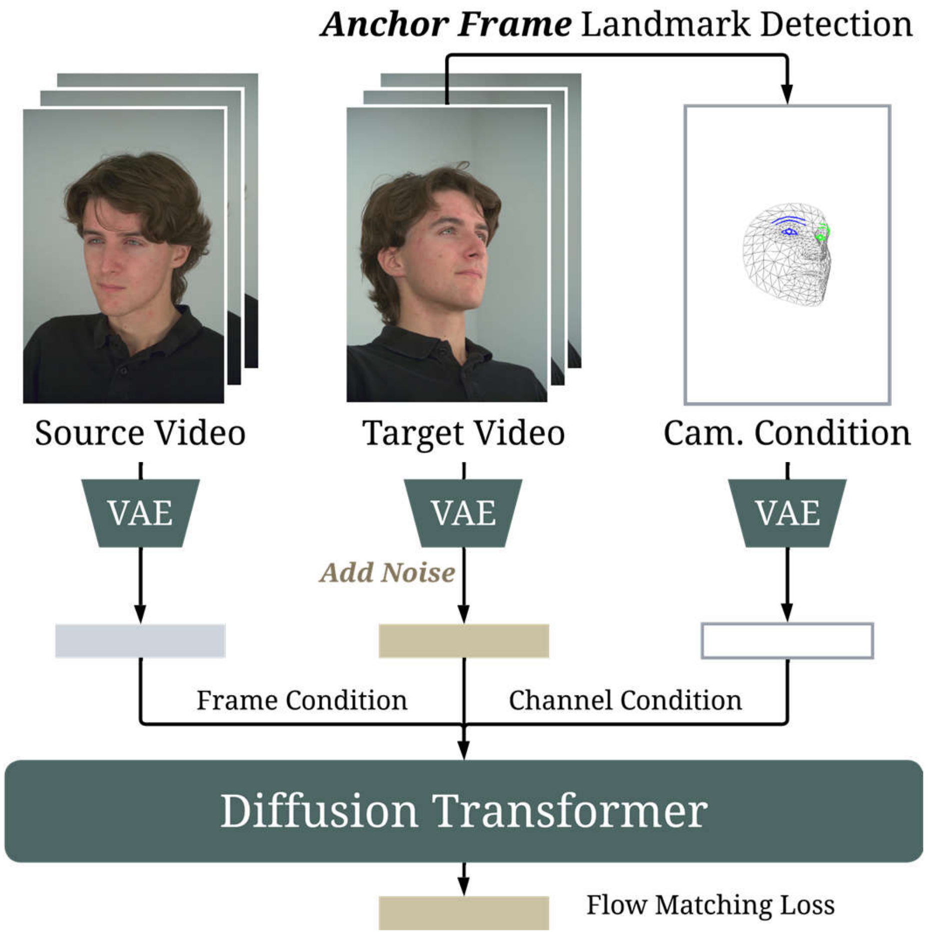
Source Video | Target Video | Source Video | Target Video



Training Pipeline

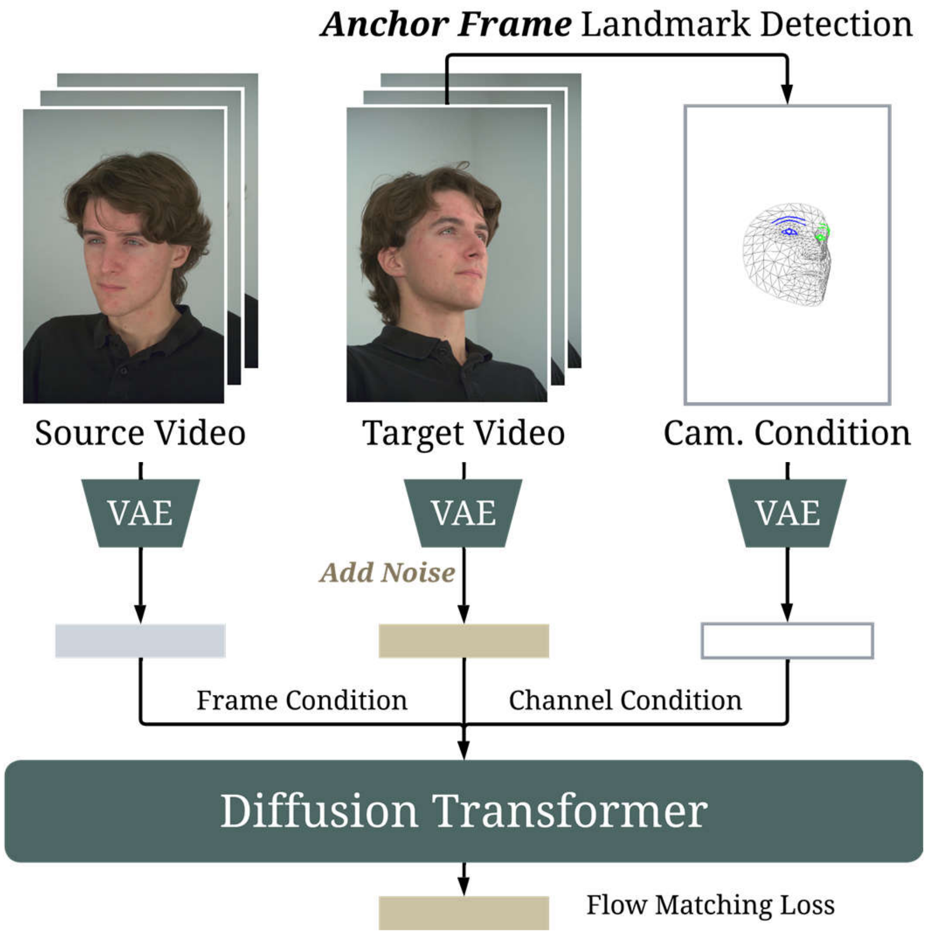
Camera Representation

- Need to disentangle head motion with the camera motion
- Extract facial landmarks from the **anchor frame (first frame)** of target video



Training Pipeline
Dual Conditioning

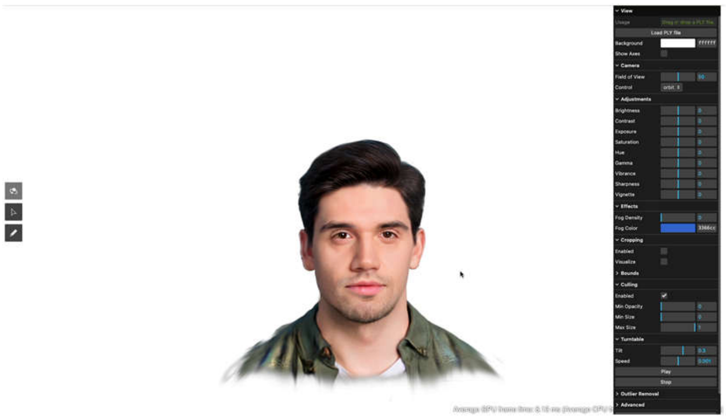
- Source video conditioned through frame conditioning, providing identity and motion context
- Target camera representation conditioned via channel conditioning, providing camera-control signal
- Similar to Wan



Inference Pipeline

- During inference, how do we get facial landmarks?
  - Render a **generic 3D head model** along the target trajectory
  - Detect facial landmarks of the rendered video as the camera condition

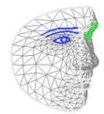
Proxy 3D Head



Proxy Video

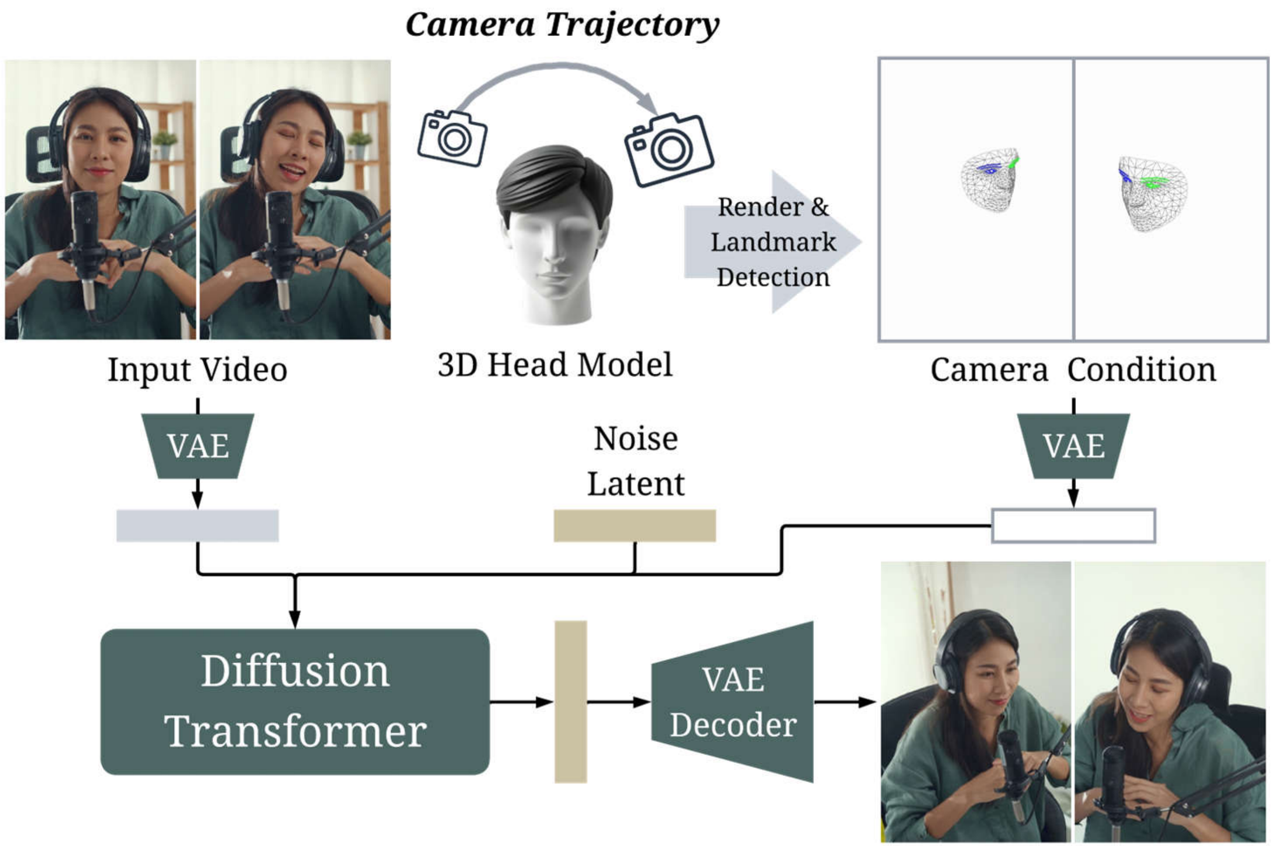
Landmark Detection

Camera Condition



Lyu, Weijie, et al. "FaceLift: Learning generalizable single image 3D face reconstruction from synthetic heads." *Proceedings of ICCV* 2025

Inference Pipeline



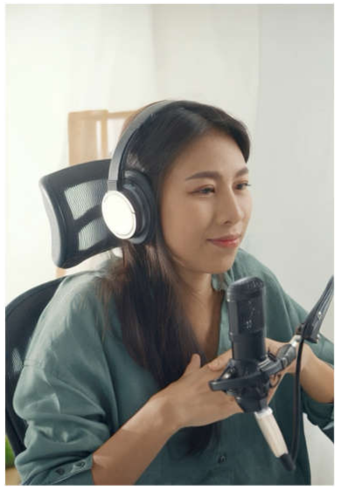
Generalization Ability on Camera Trajectories

Training | *generalize to* | Inference

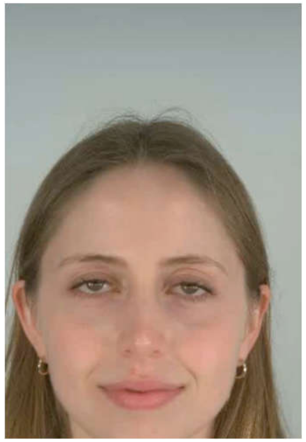
Discontinuous
Camera Pose Changes

→

Continuous
Camera Trajectories

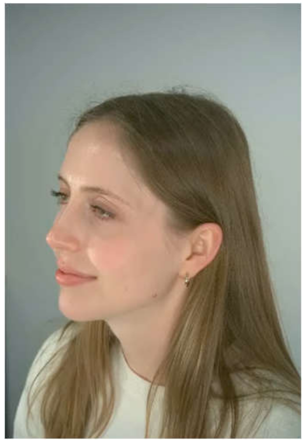


Generalization Ability on Camera Trajectories



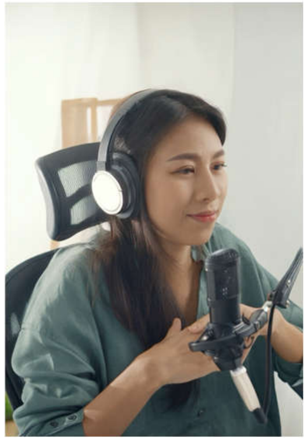
Synthetic Camera Motion

- **Continuous** camera motion
- Unchanged camera angle



Multi-shot Stitching

- Discontinuous camera motion
- **Changed** camera angles



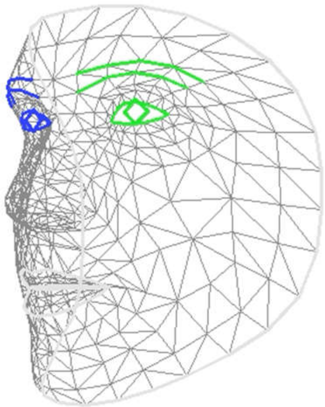
Continuously changed camera angles

Static Camera Trajectory

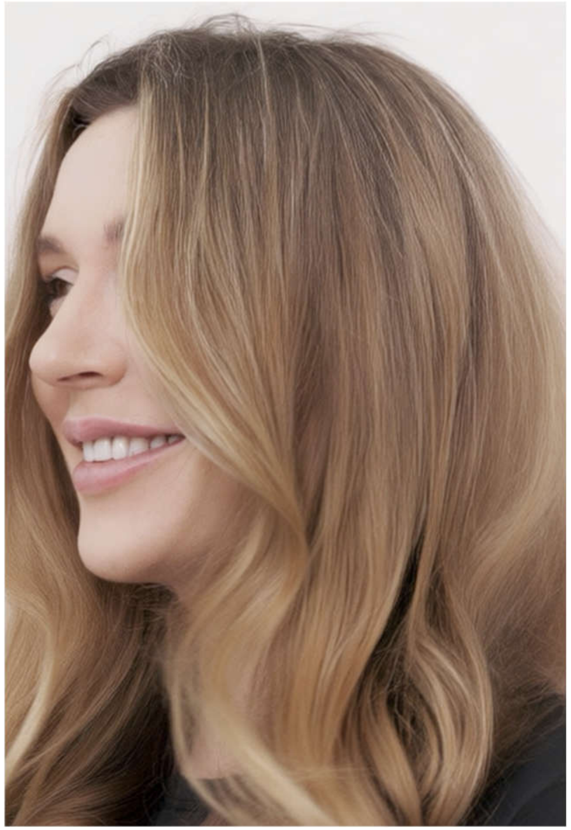
Disentanglement of Head and Camera Motion

Source Video

Target Camera

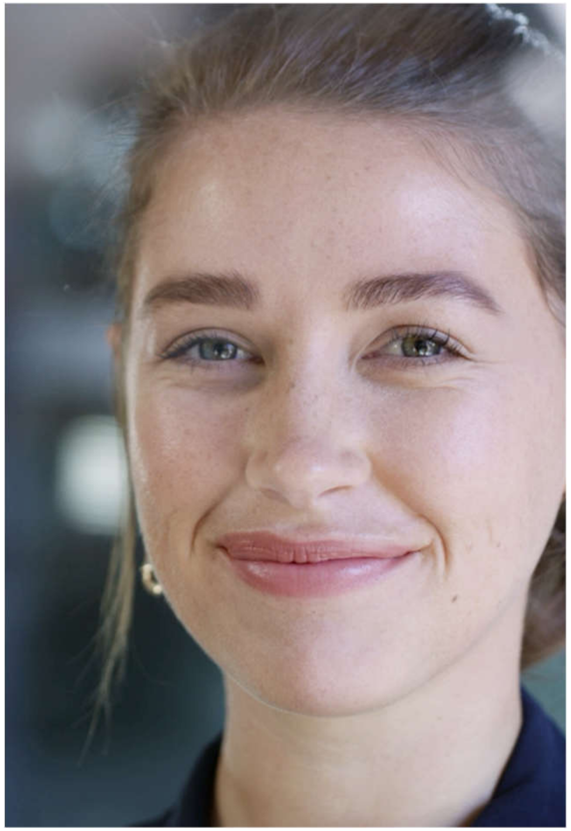


Camera Control Result

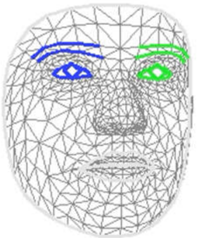


Realistic Outpainting

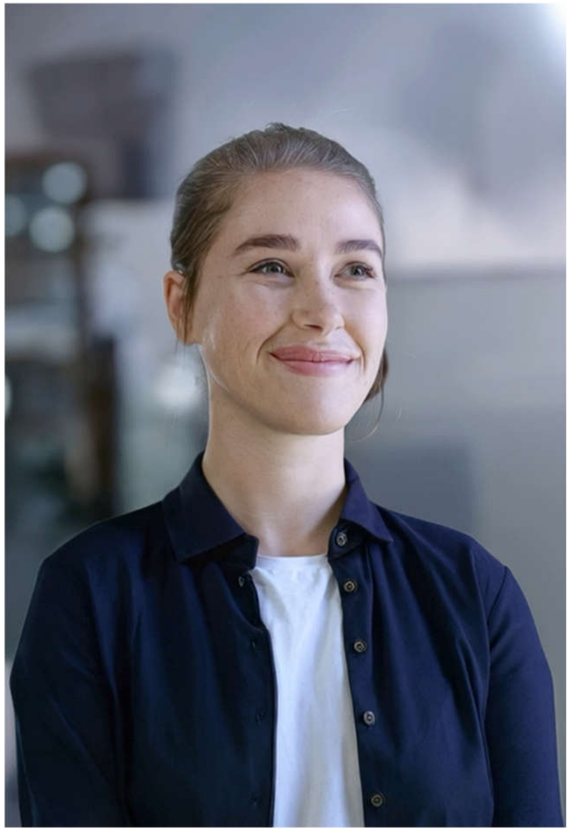
Source Video



Target Camera



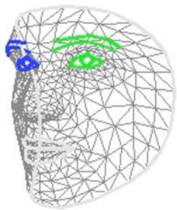
Camera Control Result



Adding a Virtual Camera

Source Video

Target Camera



Camera Control Result

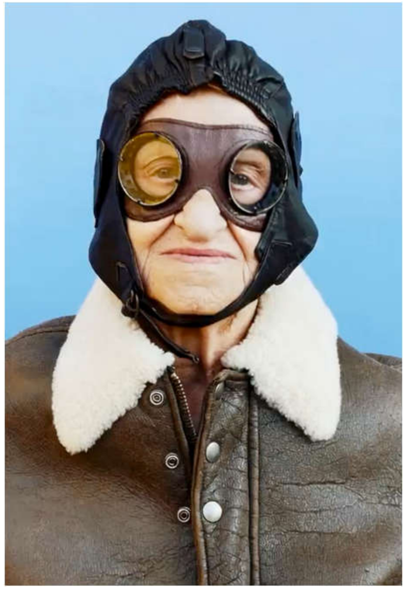
Dynamic Camera Trajectory

Comparison with Baselines

In-the-wild videos (Arc Right)

Source Video

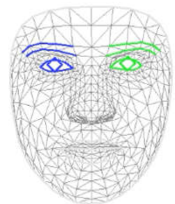
ReCamMaster



TrajectoryCrafter

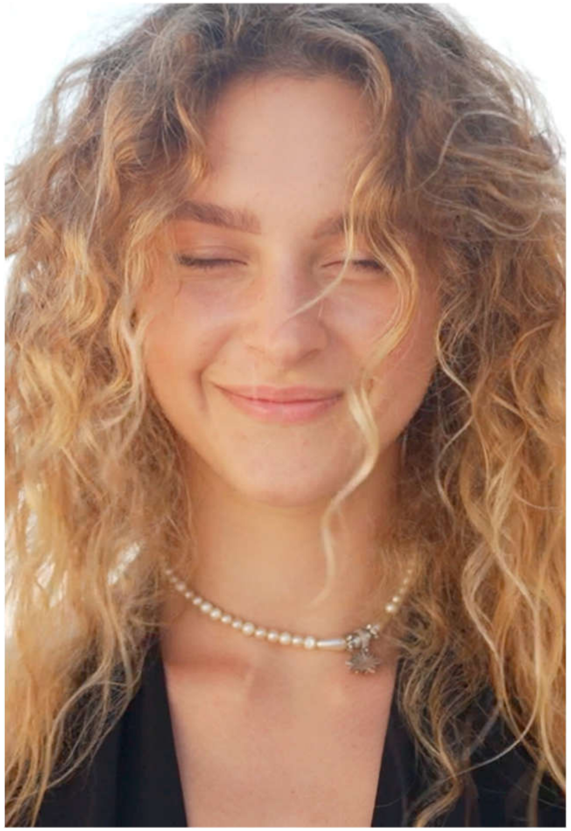
FaceCam

Target Camera

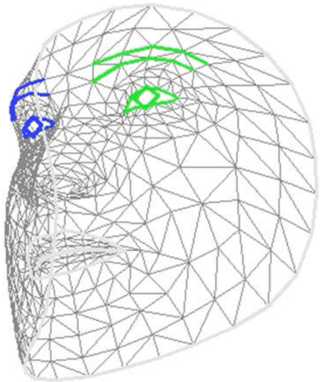


Diverse Camera Trajectories

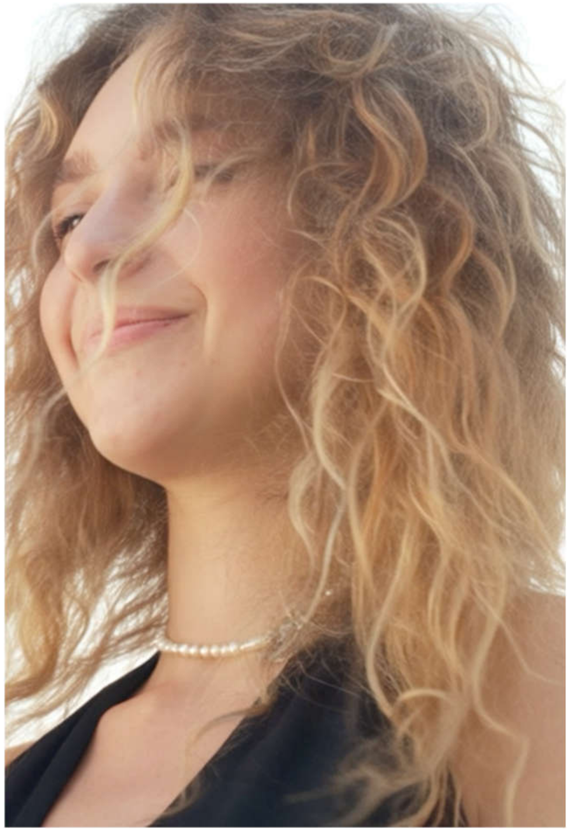
Source Video



Target Camera

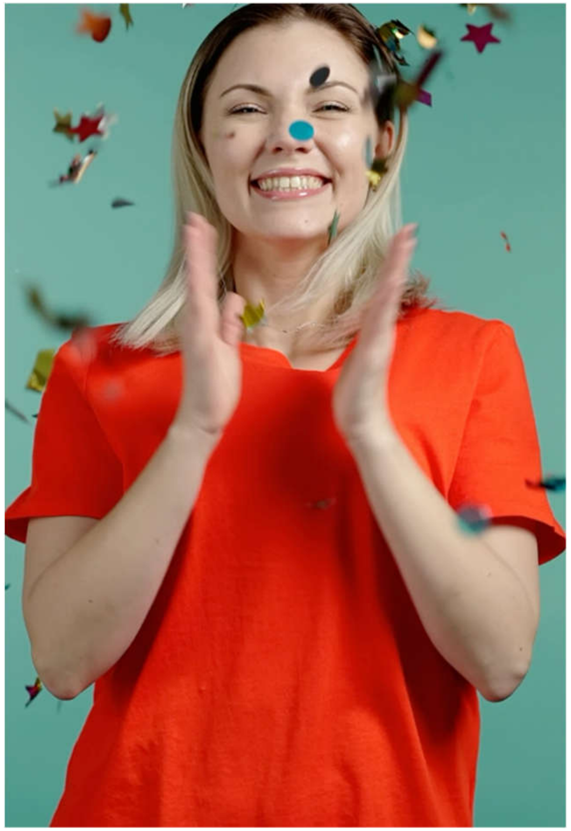


Camera Control Result

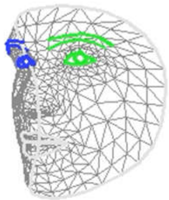


Diverse Camera Trajectories

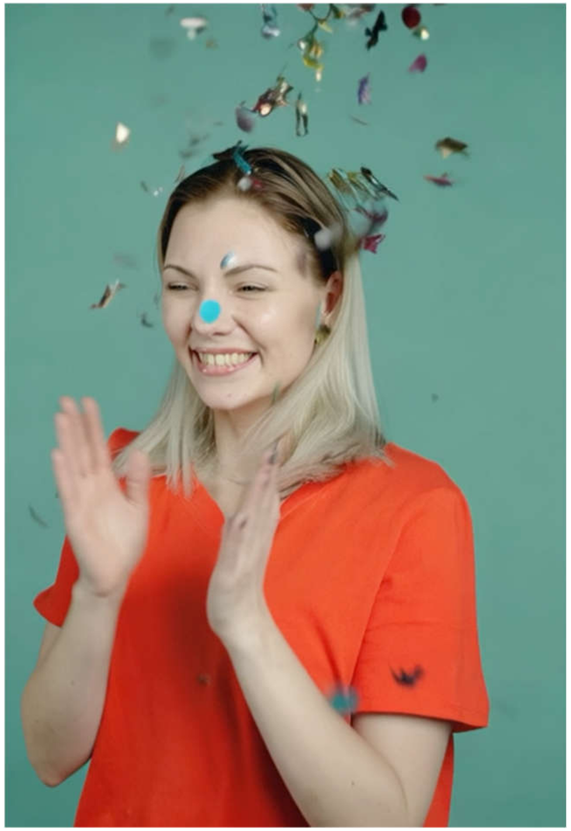
Source Video



Target Camera



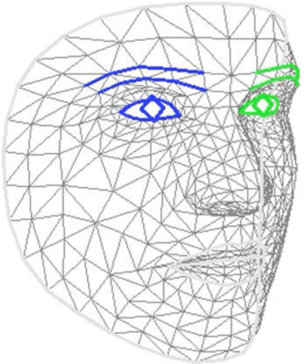
Camera Control Result



Diverse Camera Trajectories

Source Video

Target Camera

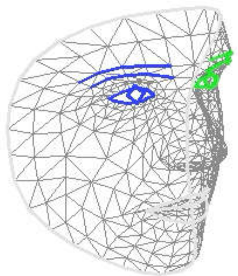


Camera Control Result

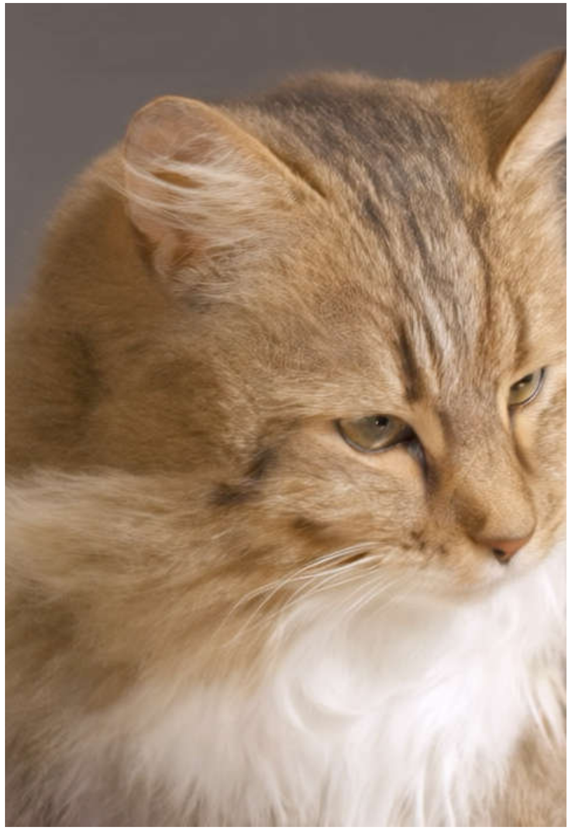
Diverse Camera Trajectories

Source Video

Target Camera



Camera Control Result



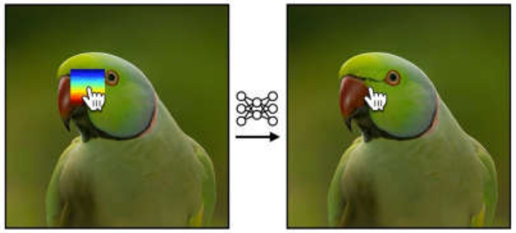
MotiMotion: Motion-Controlled Video Generation with Visual Reasoning

Lee Hsin-Ying Hanwen Jiang Yiqun Mei Jing Shi Ming-Hsuan Yang Zhixin Shu
University of California, Merced Adobe Research

ICML 2026

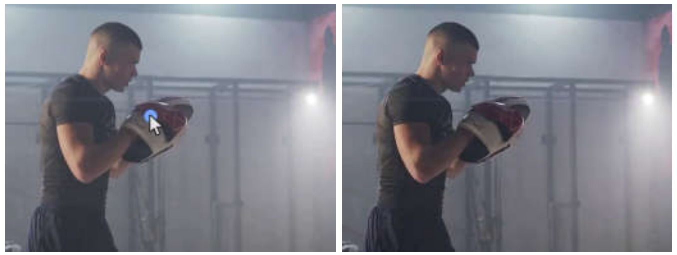
Controlling Motion

User trajectories are often sparse and indicate only primary motion.



Motion Prompting

Tora



WanMove

→ What if the motion trigger an event or affect the environment?

Modeling Reactive and Secondary Motion

Motion should align with physics and common-sense knowledge.

Collision | Constraint Change | Airflow | Tool Mechanism | Common Objects

→ Current video generators can still fail to model such dynamics.

Aligning with Physics and Common Sense

A VLM predicts what may happen and how the objects move

User Trajectories

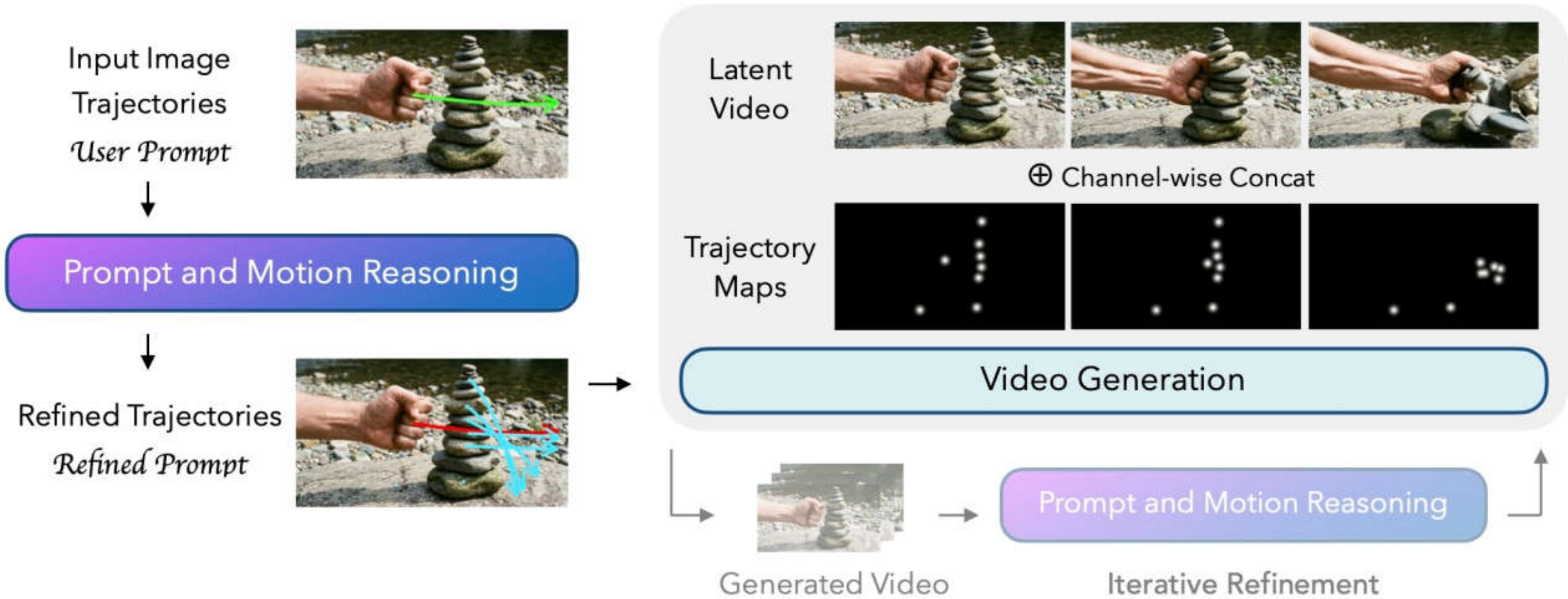
Predicted Trajectories

Red: refined user trajectories
Blue: proposed new trajectories

Video

Reasoning-then-Generation

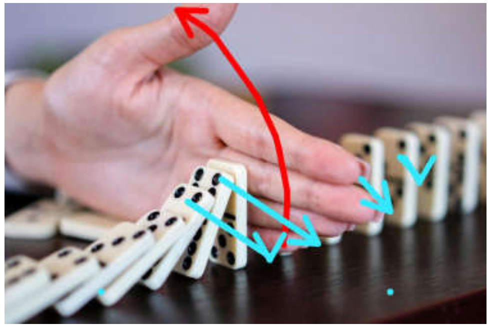
VLM proposes a plan and then video generator executes the plan

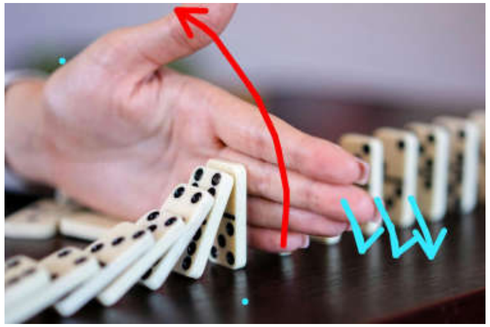


Imperfect Trajectories

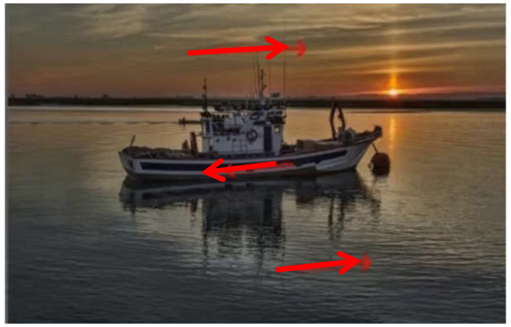
VLM predictions sometimes don't align perfectly.

Human trajectories are often smooth and simple.





Shifted



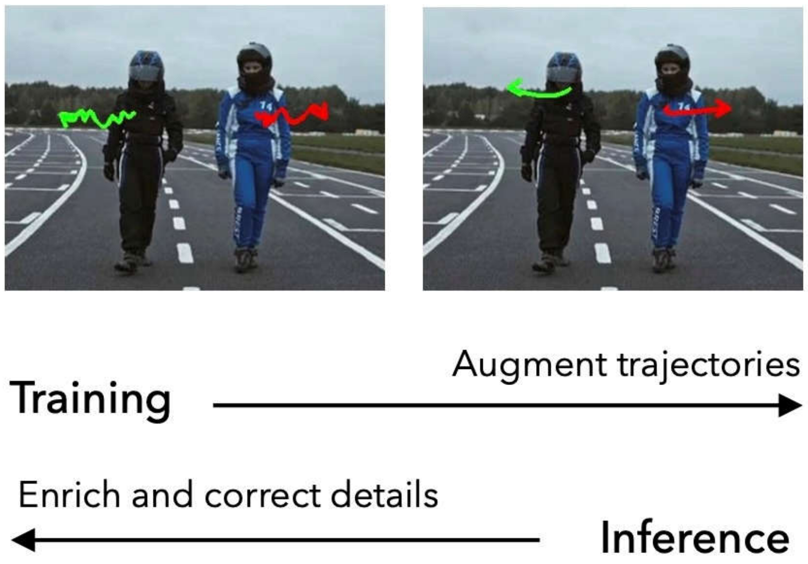
Human Annotation
(MotionPro)

Tracking
(MoveBench)

Confidence-Aware Control

Learning to follow trajectories elastically according to confidence

- High confidence: follow strictly
- Low confidence: allow the video generator to improvise → scale Gaussian maps in trajectory control



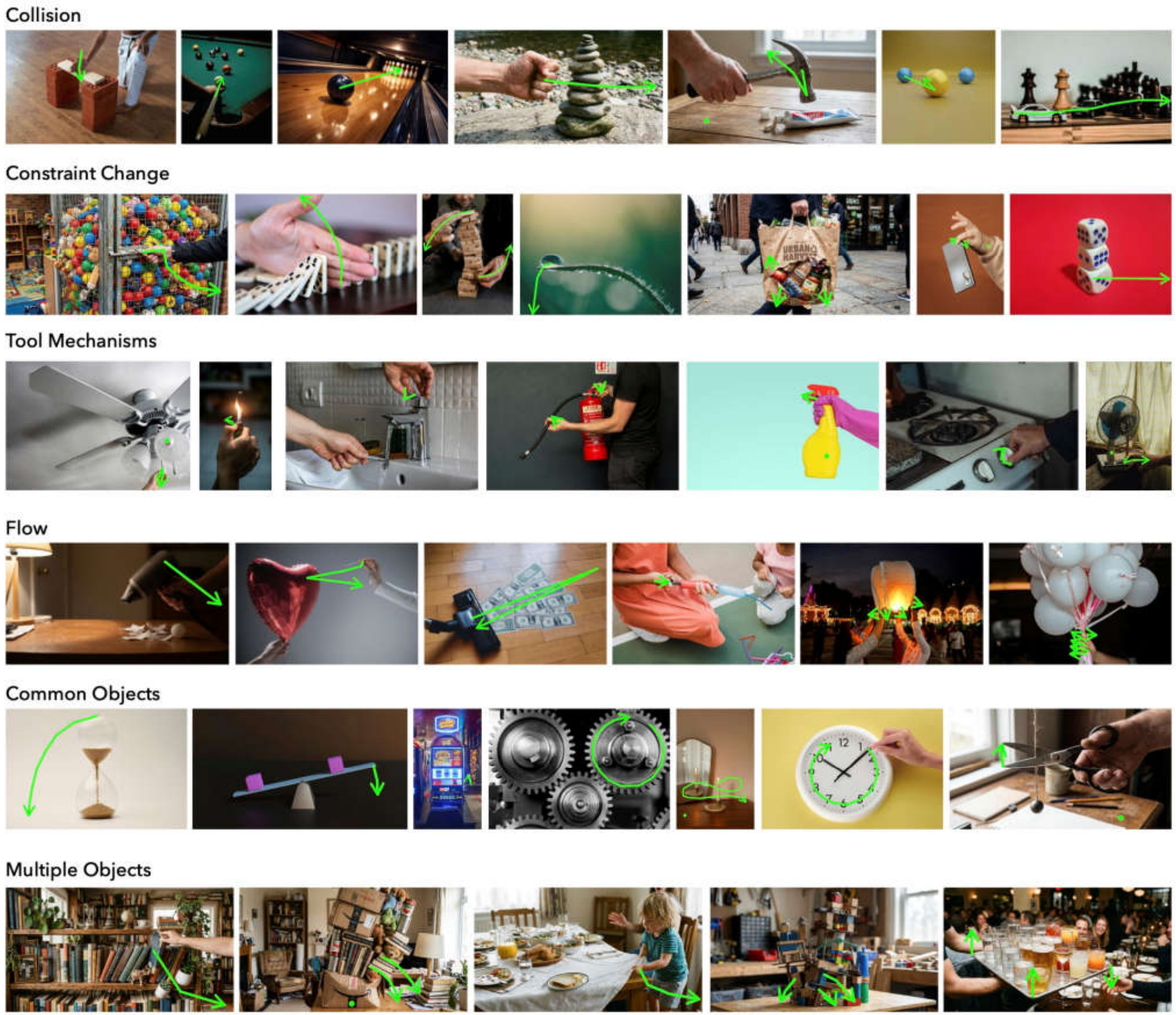
MotiBench

Pre-event scenes

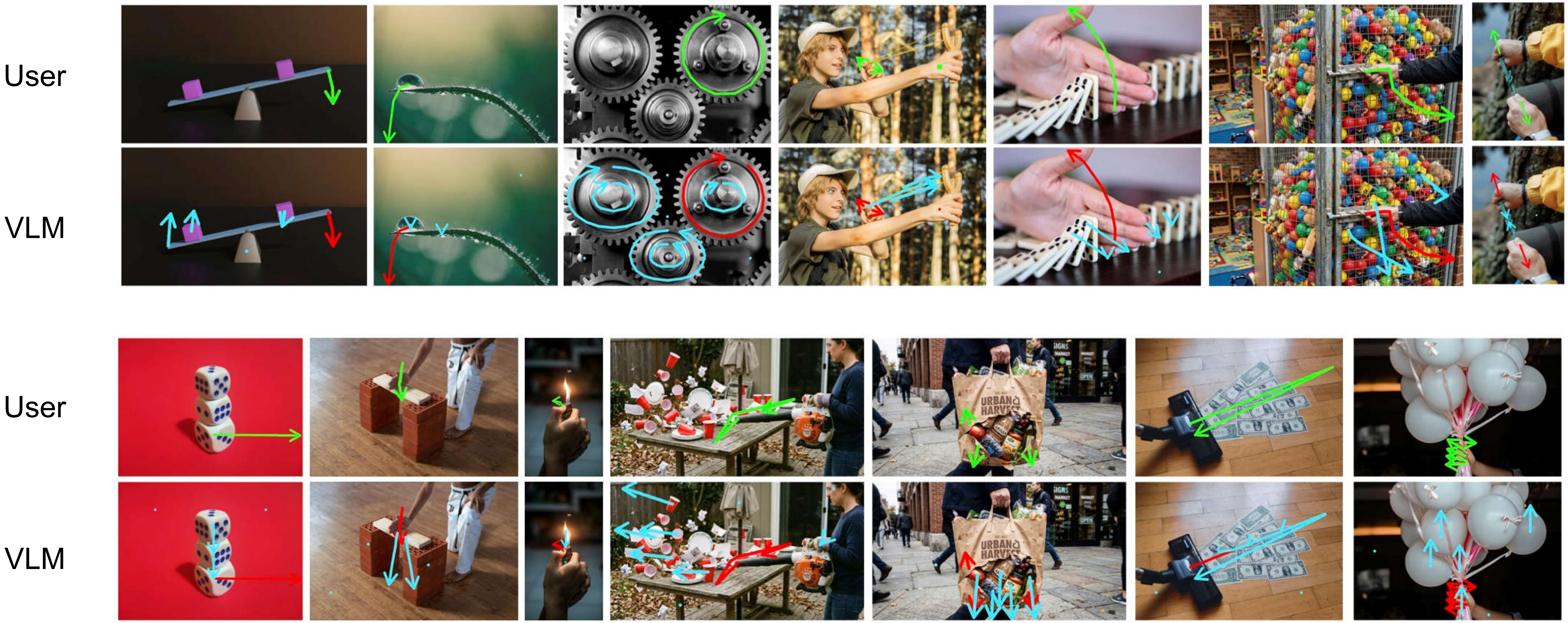
Images, trajectories, and prompts

Category	Num of Samples	Percentage (%)
Collision	9	15
Constraint Change	17	27
Tool Mechanisms	8	13
Flow	9	14
Common Objects	19	31

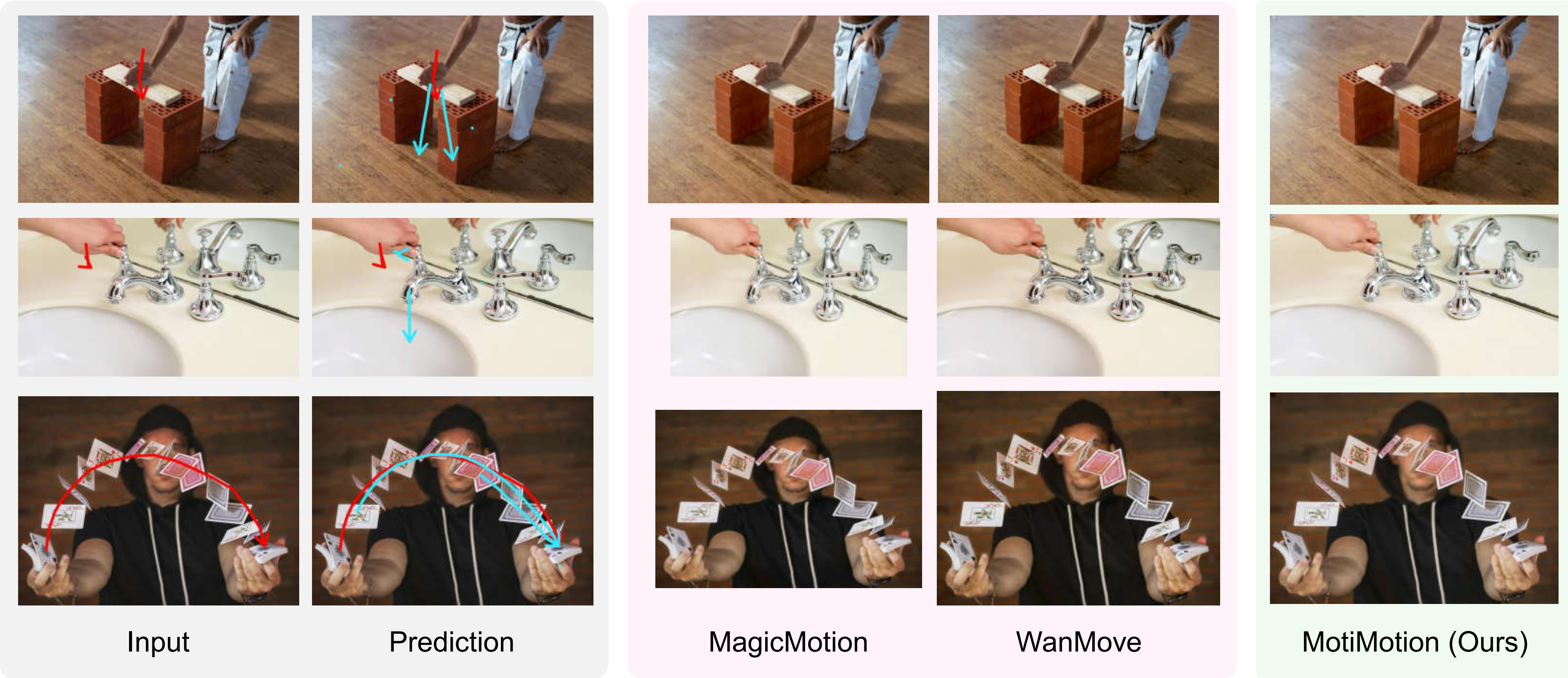
Statistics



VLM Prediction

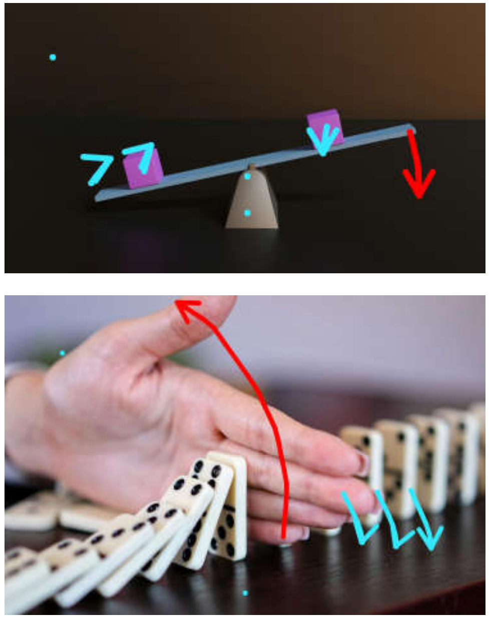


Comparison

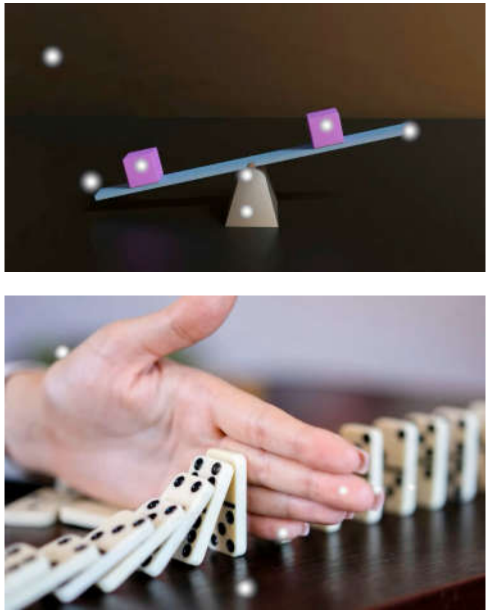


More results in https://motimotion.github.io

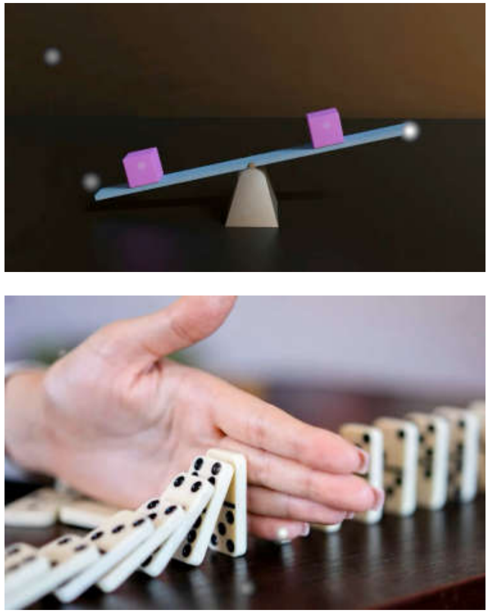
Ablation Study



Prediction



High Confidence



Low Confidence

Summary

- Identify gap in motion-controlled video generation and address it with VLMs.
- MotiMotion: VLM-driven semantic and motion control, with confidence modulation for natural motion.
- MotiBench: a benchmark for physical and causal reasoning, where MotiMotion is preferred over prior methods.

Context Forcing: Consistent Autoregressive Video Generation with Long Context

Shuo Chen Cong Wei Sun Sun Tiancheng Shen Ping Nie Kai Zou Ge Zhang
Ming-Hsuan Yang Wenhu Chen

ICML 2026

Causal Autoregressive Video Generation

Causal autoregressive video generation

CausVid (Teacher Forcing)

- Distill slow bi-directional teacher model to few-step autoregressive model

Self Forcing: Bridge training and inference gap

- Student self rollout, using bi-directional teacher model give whole video DMD loss
- Trained on short clip video

Causal autoregressive **long** video generation

LongLive & Self Forcing ++

- Student self **rollout long** sequence, using bi-directional **short teacher** model give whole video DMD loss

InfinateRoPE

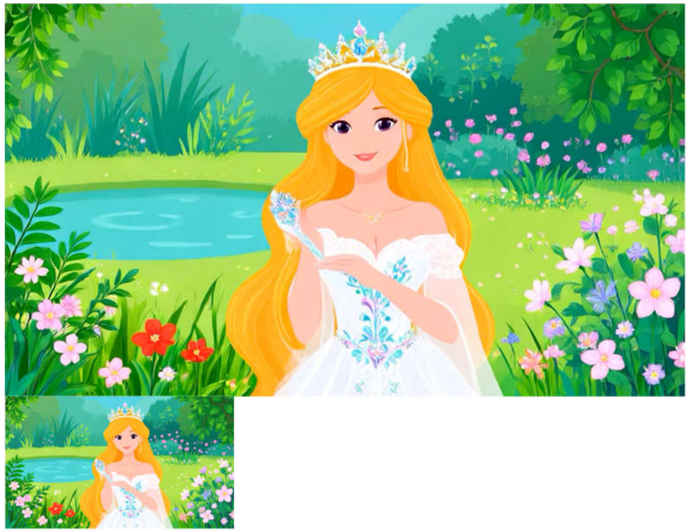
- Reduction of error accumulation by modifying positional encodings to relative ones

Forgetting Drifting Dilemma in causal long video generation

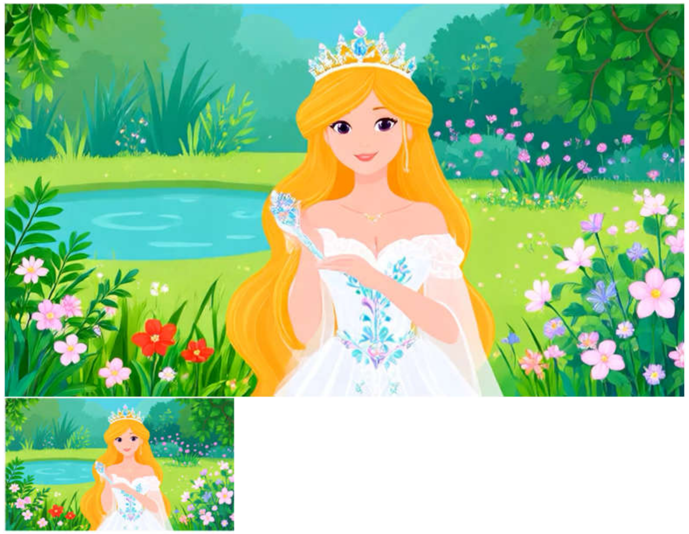
Forgetting: Restricting the model to a short memory window minimizes error accumulation, but causes the model to lose track of previous subjects and scenes during long rollout.

Drifting: Maintaining a long context preserves more previous information, but exposes the model to more errors. The video distribution progressively drifts away from the real manifold.

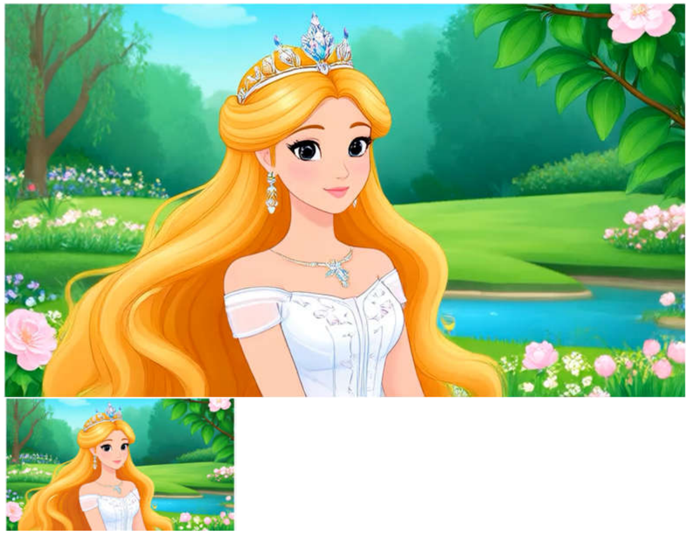
LongLive, 3.0s context (x5)



LongLive, 5.25s context (x5)

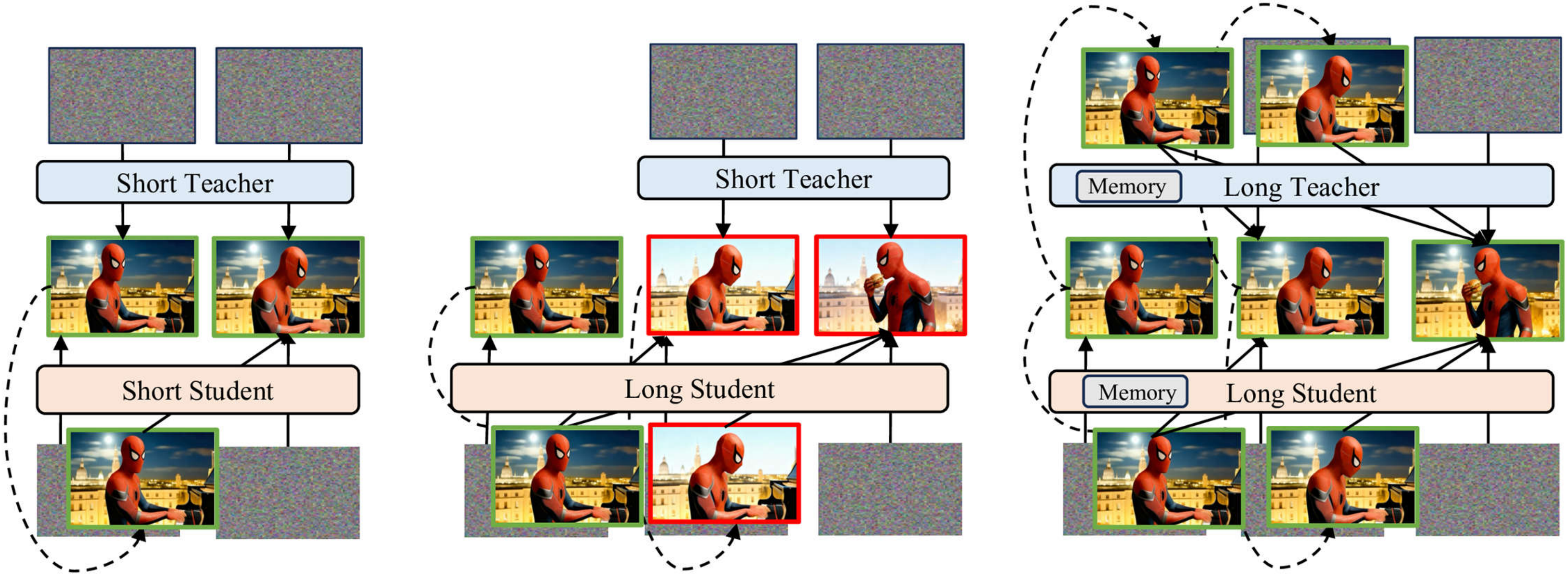


Ours, 20s+ context (x5)



[1] Yang, S., Huang, W., Chu, R., Xiao, Y., Zhao, Y., Wang, X., ... & Chen, Y. (2025). Longlive: Real-time interactive long video generation. *arXiv preprint arXiv:2509.22622*.

Context Long Tuning: Bridge Teacher-Student Mismatch



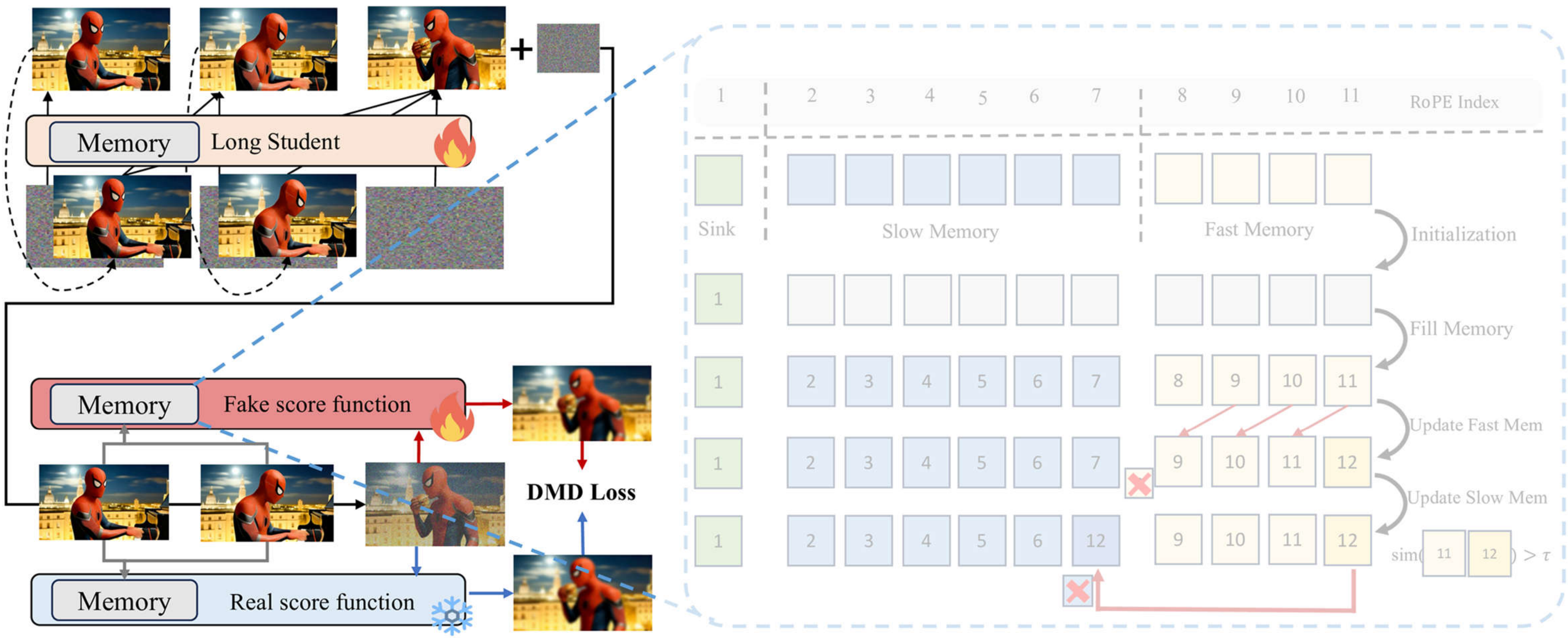
(a) Self-Forcing Training
short teacher → short student

(b) Memoryless Long Tuning
short teacher → long student

(c) **Context Long Tuning(ours)**
long teacher → long student

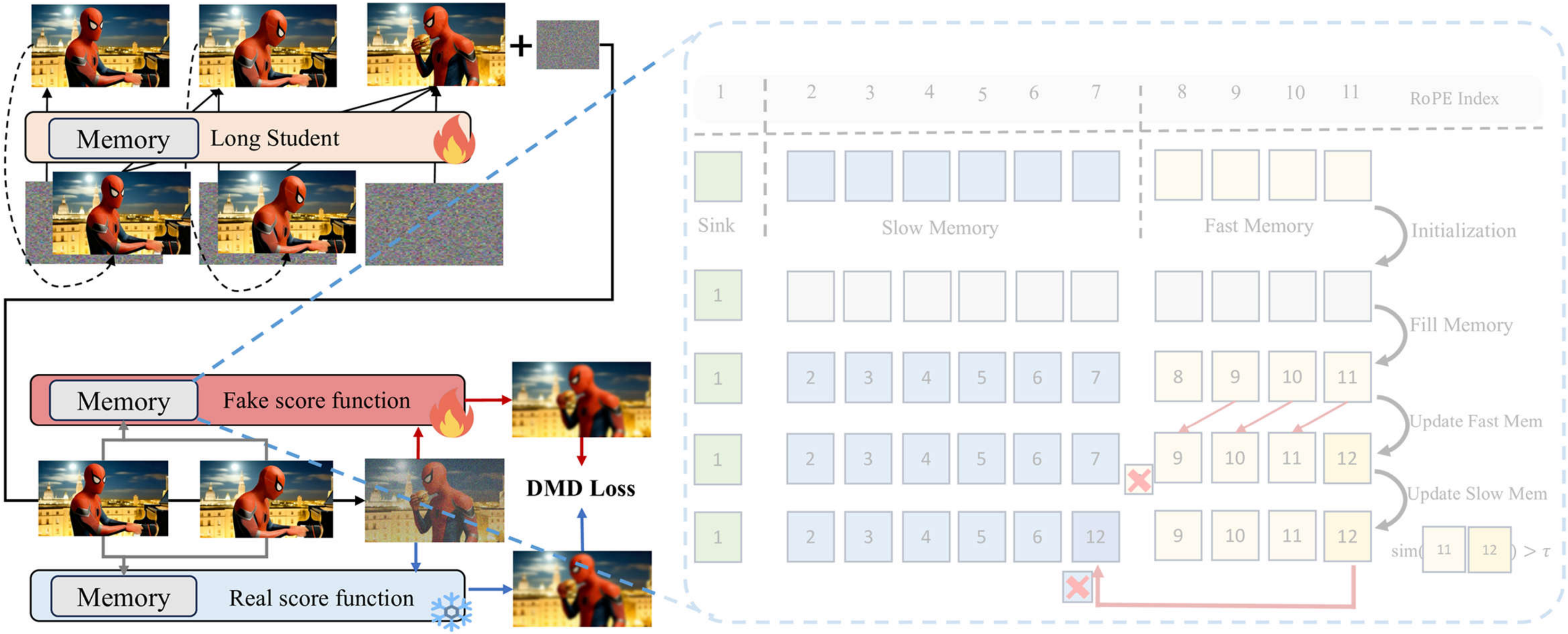
Context Forcing: The student is supervised by a long-context teacher aware of the full generation history, resolving the mismatch in (b).

Contextual Distribution Matching Distillation



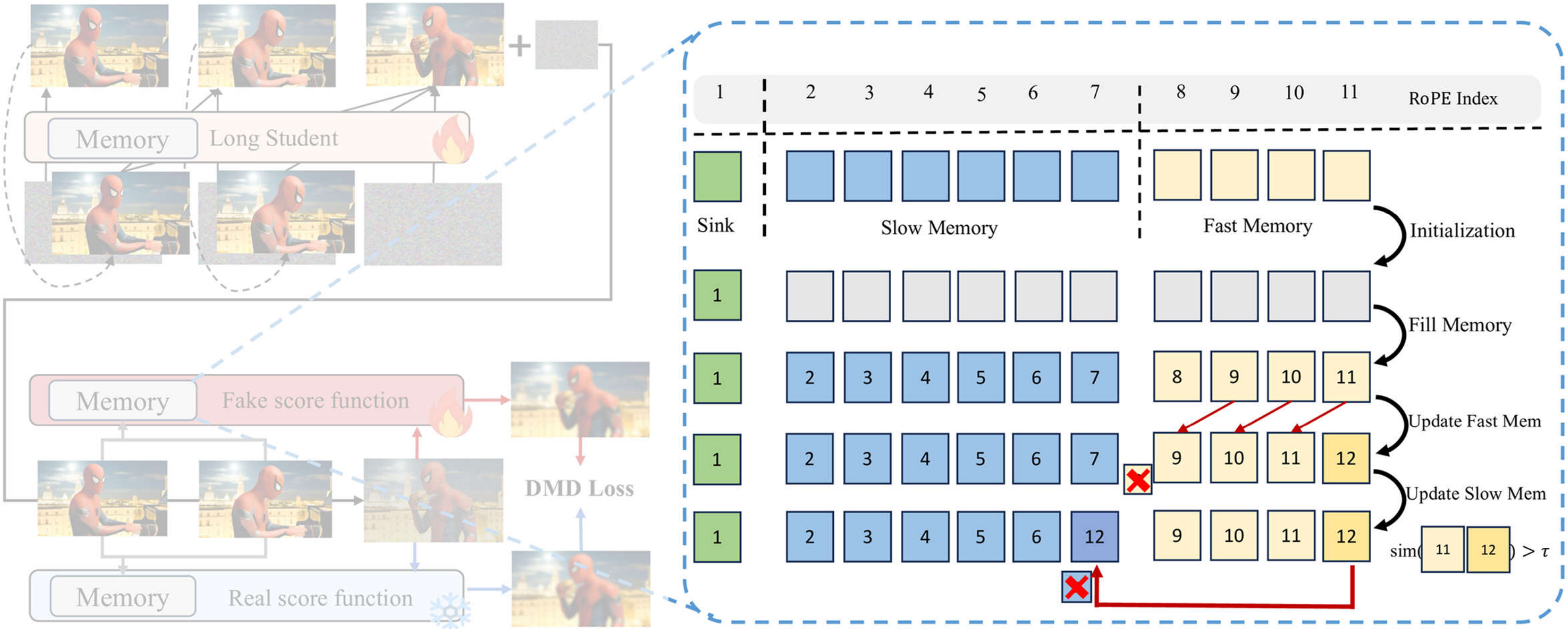
Context Forcing: Student supervised by long-context teacher aware of full generation history, resolving teacher-student mismatch
During contextual DMD training, long teacher provides supervision to long student by using the same context memory mechanism

Context Distribution Matching Distillation



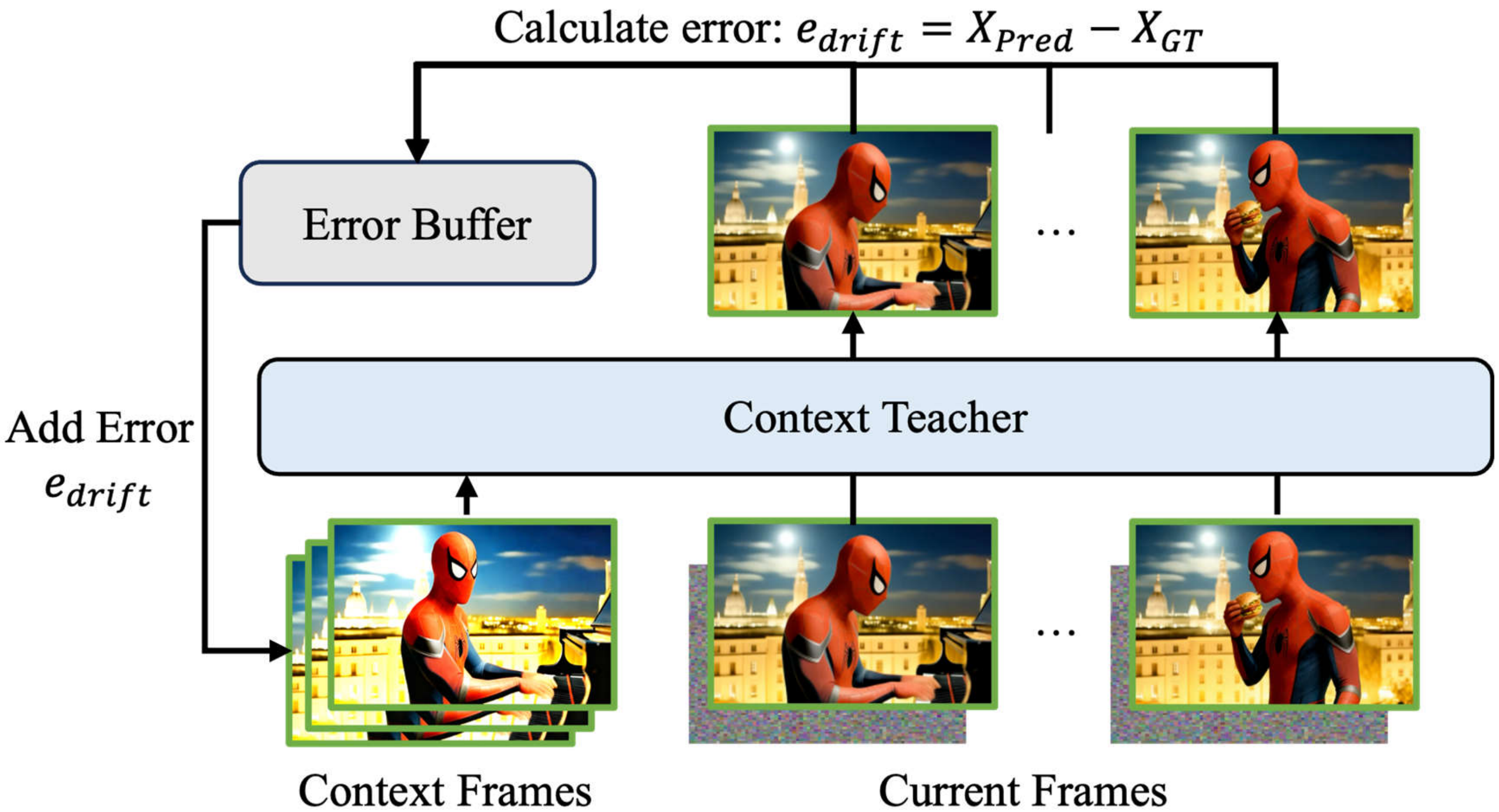
Real Score: frozen context teacher that take clean context images and noise images as input and do denoise.
Fake score: same arch with real score, learning the student distribution.
Student: autoregressive student with context management system

Context Management System



Use KV Cache as the context memory: 1). sink, 2). slow memory, 3). fast memory.
Attention Sink: retain initial tokens to stabilize attention (similar to StreamingLLM)
Slow Memory: store high-entropy keyframes and updating only with significant new information
Fast Memory: capture immediate local context with short-term persistence

Context Teacher Training

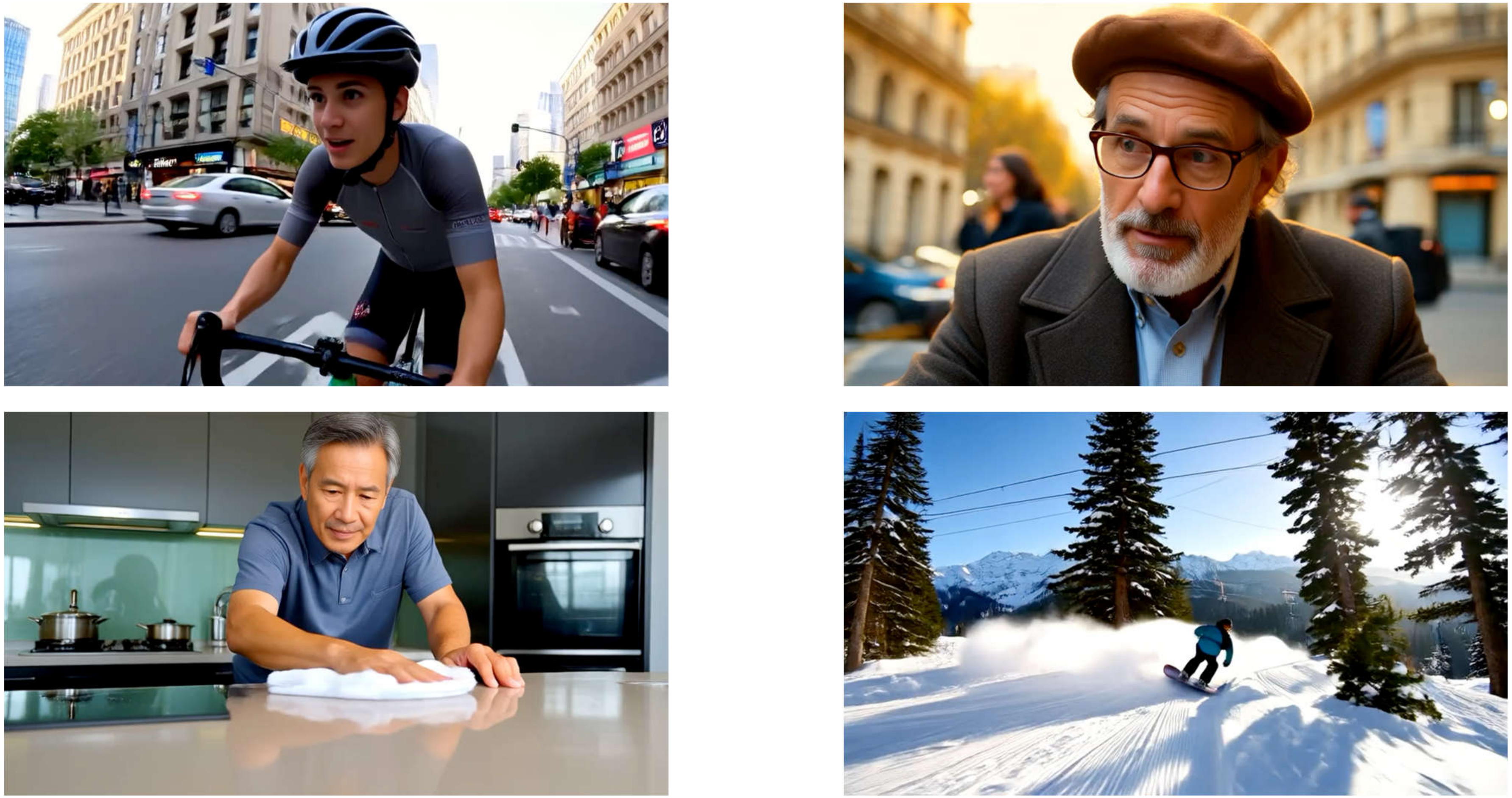


To handle exposure bias, past predictive errors are dynamically stored in an Error Buffer and probabilistically **injected into clean context and noise latents** (similar to error recycling fine tuning)

The input sequence aligns directly with the student model's **context management system** (incorporating 21 latent frames of sink, slow, and fast memory).

Long Video Generation

Qualitative Results of Context Forcing. Our method enables minute-level video generation with minimal drifting and high consistency across diverse scenarios.



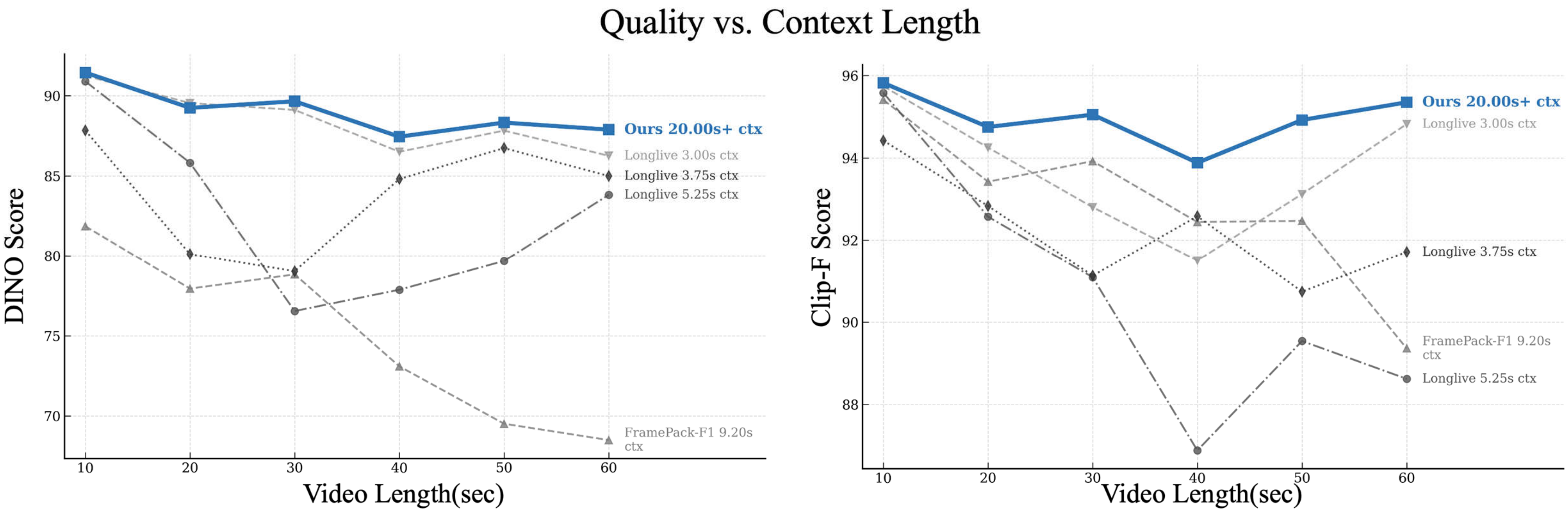
Long Video Generation

Qualitative Results of Context Forcing. Our method enables minute-level video generation with minimal drifting and high consistency across diverse scenarios.

Consistency Evaluation

For streaming long-context tuning baselines (e.g., LongLive),enlarging the context window during inference (3.0 → 5.25 s) causes error accumulation and distribution shift (Drifting).

Context Forcing supports 20s+ context while maintaining strong long-term consistency.



Consistency Evaluation

Our method maintains both the background and subject consistent across 1-min video, while other baselines have different levels drifting or identity shift.

LongLive[1]
(KV Size = 12)

Infinite Rope[2]
(KV Size = 6)

Ours(KV Size = 21)

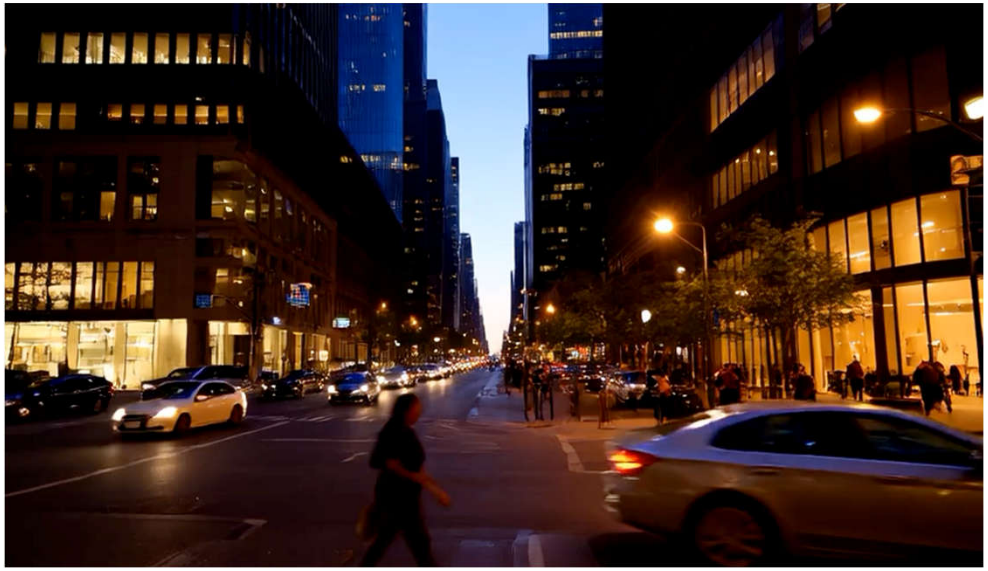
[1] Yang, S., Huang, W., Chu, R., Xiao, Y., Zhao, Y., Wang, X., ... & Chen, Y. (2025). Longlive: Real-time interactive long video generation. *arXiv preprint arXiv:2509.22622*.
[2] Yesiltepe, H., Meral, T. H. S., Akan, A. K., Oktay, K., & Yanardag, P. (2025). Infinity-RoPE: Action-Controllable Infinite Video Generation Emerges From Autoregressive Self-Rollout. arXiv preprint arXiv:2511.20649.

Comparison with LongLive

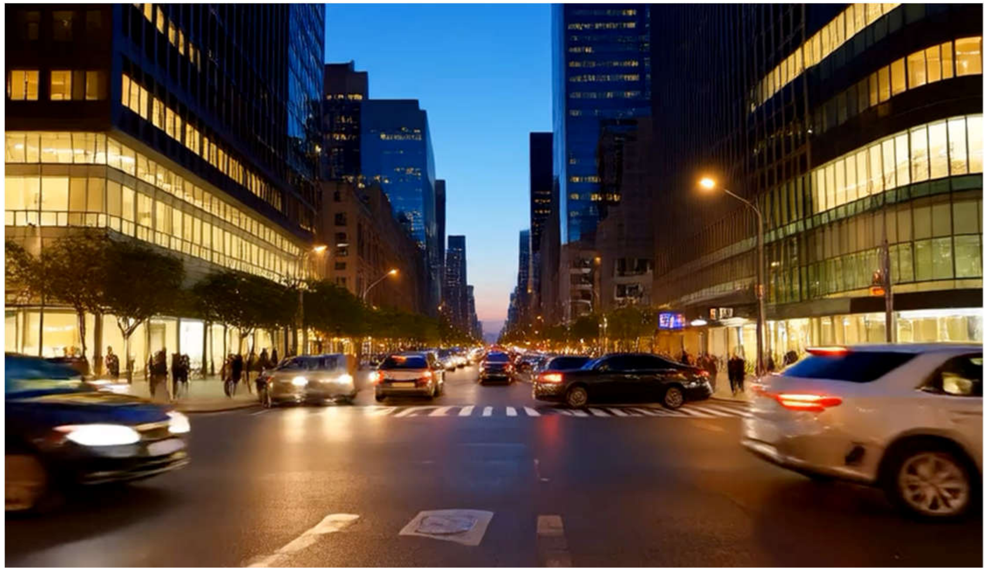
Context Forcing enables stable **minute-level video generation** with high **subject/background consistency** across diverse scenarios.

LongLive exhibits **flashback artifacts**(e.g., at 50s) and lacks the capacity to preserve consistent subject and background details throughout long sequences.

Longlive



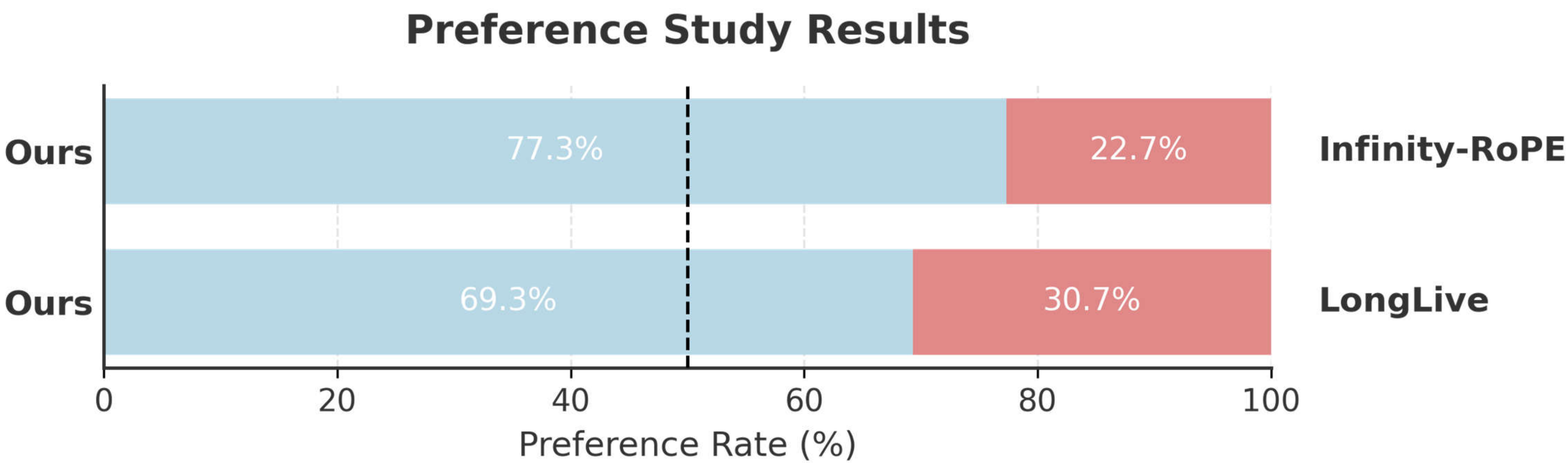
Ours



[1] Yang, S., Huang, W., Chu, R., Xiao, Y., Zhao, Y., Wang, X., ... & Chen, Y. (2025). Longlive: Real-time interactive long video generation. *arXiv preprint arXiv:2509.22622*.

User Study

Human raters compare two anonymized videos and indicate which is more aesthetic, better matches the prompt, and exhibits stronger temporal consistency (subject and background coherence).



Blind preference rates (higher is better for Ours): vs. Infinity-RoPE (77.3% / 22.7%) and vs. LongLive (69.3% / 30.7%). The dashed line marks 50% (chance).

Ablation Study

Ablation study on Slow Memory Sampling Strategy, Context DMD, and Bounded Positional Encoding (evaluated on 60s).

Our Context DMD and Bounded RoPE increase the long video consistency and mitigating temporal drift during the generation process.

Model	Total Score ↑	Quality Score ↑	Semantic Score ↑	Background Consistency ↑	Subject Consistency ↑	Dynamic Degree ↑
Slow Memory Sampling Strategy						
Uniform sample, interval 1	80.82	82.20	<u>75.32</u>	92.45	92.10	52.15
Uniform sample, interval 2	<u>81.11</u>	<u>82.61</u>	75.12	93.12	92.85	<u>55.30</u>
Contextual Distillation						
w/o. Contextual Distillation	80.36	82.28	72.70	<u>93.55</u>	<u>93.20</u>	48.12
Bounded Positional Encoding						
w/o. Bounded Positional Encoding	73.52	75.44	65.82	84.68	79.24	27.45
Ours	**82.45**	**83.55**	**76.10**	**95.34**	**94.88**	**58.26**

Concluding Remarks

- We propose **Context Forcing**, a framework that enables **consistent long-video generation** by aligning student and teacher context lengths.

- Paired with a **Slow-Fast Memory system**, it achieves 2--10x **longer context information** than current state-of-the-art methods.

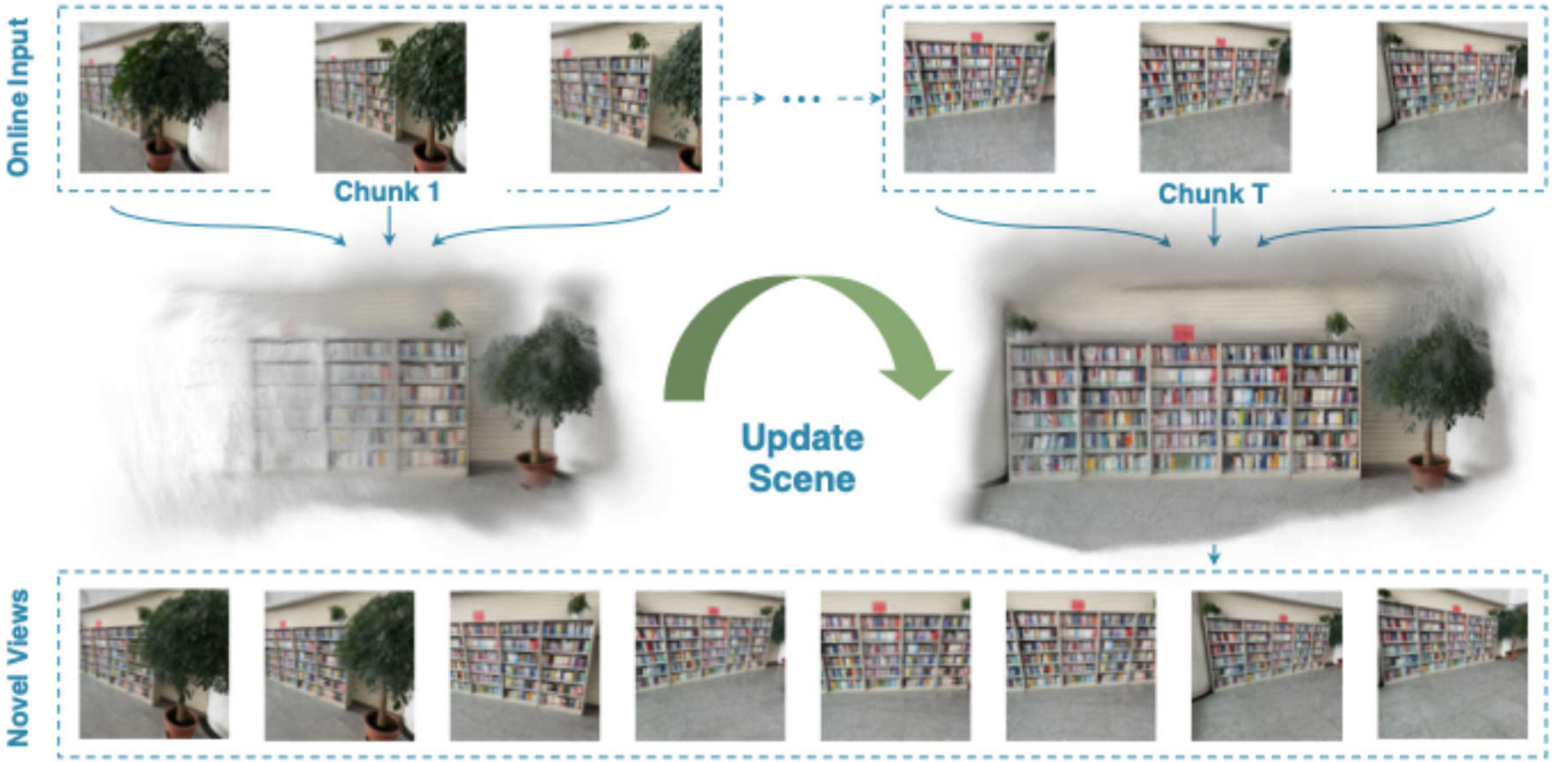
Future Work

- Learnable Memory? (FramePack, PFP, etc)
- User Input? (Genie 3, LingBot-World)
- Infra? (FastVideo, LongLive-NVFP4)

ReCoSplat: Autoregressive Feed-Forward Gaussian Splatting Using Render-and-Compare

ReCoSplat

How to construct an explicit 3D representation of the world that can be updated with new observations?

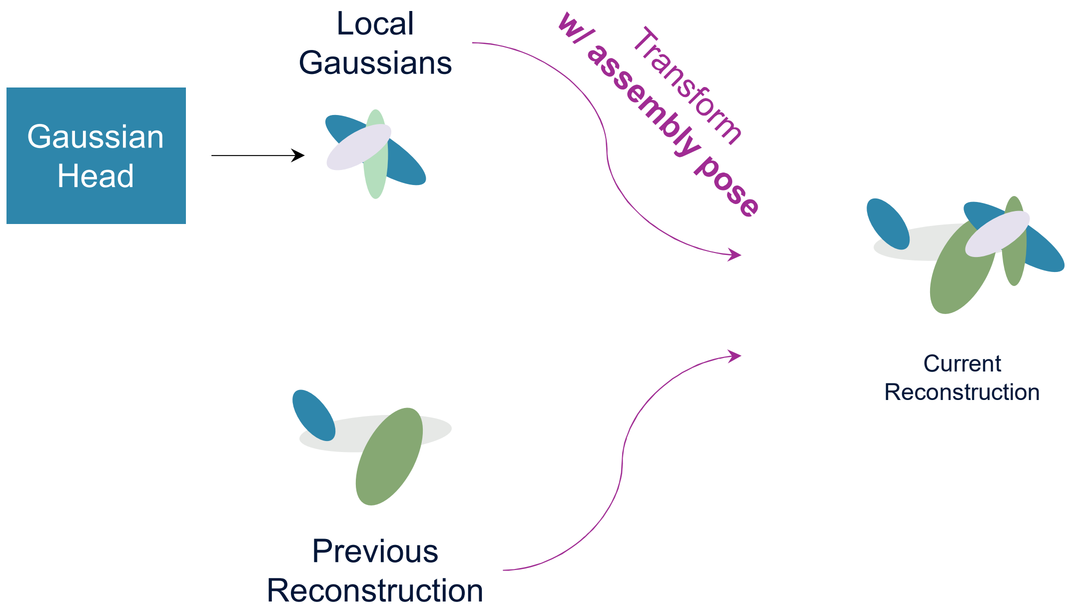


ReCoSplat

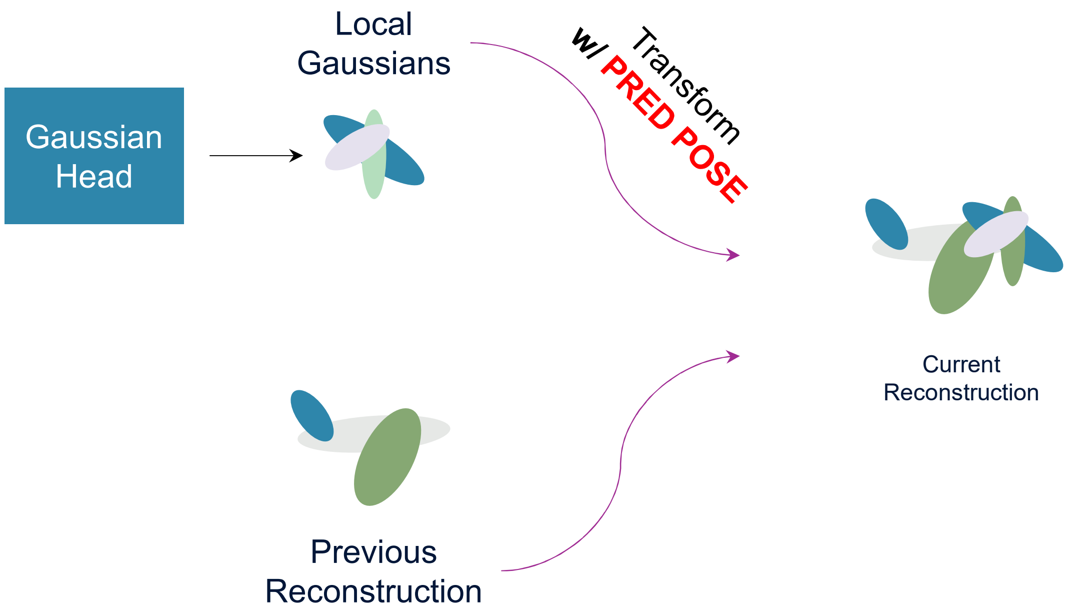
- Optimization-based: lengthy per-scene training.
- Feed-forward: offline reconstruction.
- Autoregressive setting?
- Supporting both posed and unposed inputs?

Assembly Pose

- Local Gaussian prediction > canonical space prediction.
- But requires local-to-world coordinate transformation with camera extrinsics.
- Call the camera poses used for this mapping **“assembly poses”**.

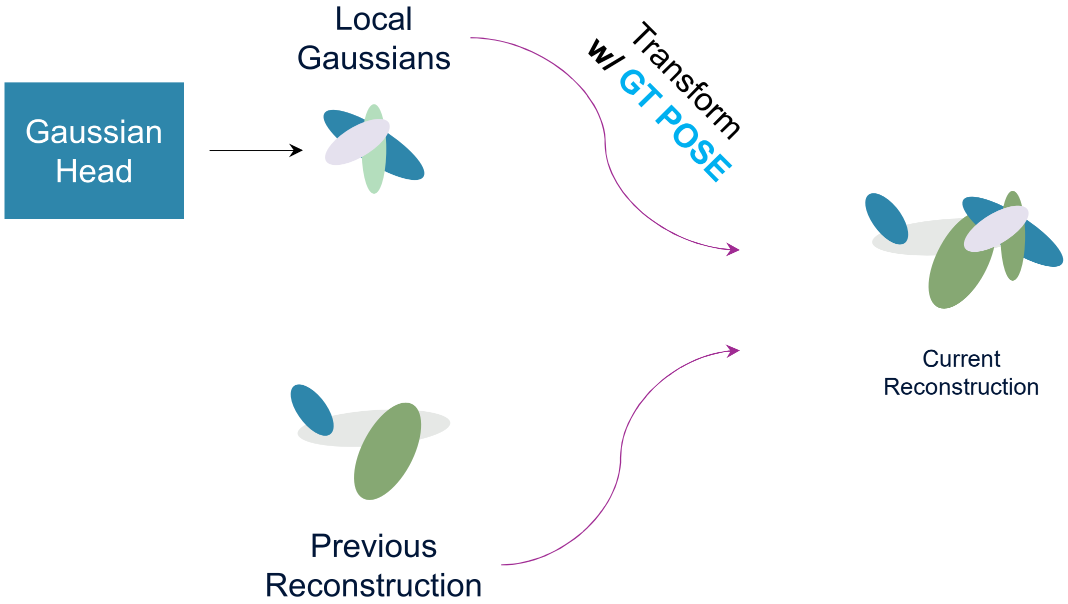


Pose Distribution Mismatch



Method	Stage	Posed?	k	PSNR ↑	SSIM ↑	LPIPS ↓
KV Cache	1		✓	21.346	0.680	0.224
KV Cache w/ Predicted Assembly Pose	1		✓	20.781	0.641	0.238
KV Cache	1	✓	✓	21.866	0.729	0.196
KV Cache w/ Predicted Assembly Pose	1	✓	✓	20.836	0.656	0.229

Pose Distribution Mismatch



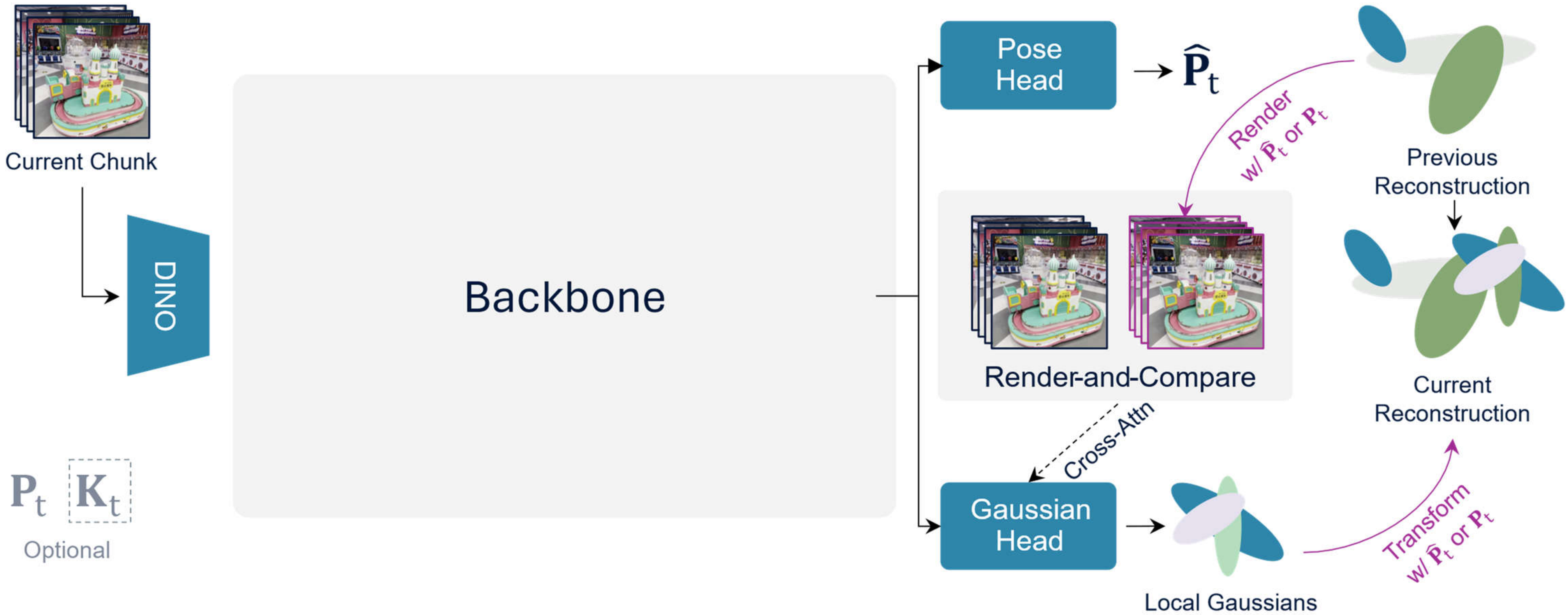
Method	Stage	Posed?	k	PSNR ↑	SSIM ↑	LPIPS ↓
KV Cache	1		✓	21.346	0.680	0.224
KV Cache	1	✓	✓	21.866	0.729	0.196

Solution: Render-and-Compare Module

Naively use
GT assembly:
train-test mismatch

Ours

Solution: Render-and-Compare Module

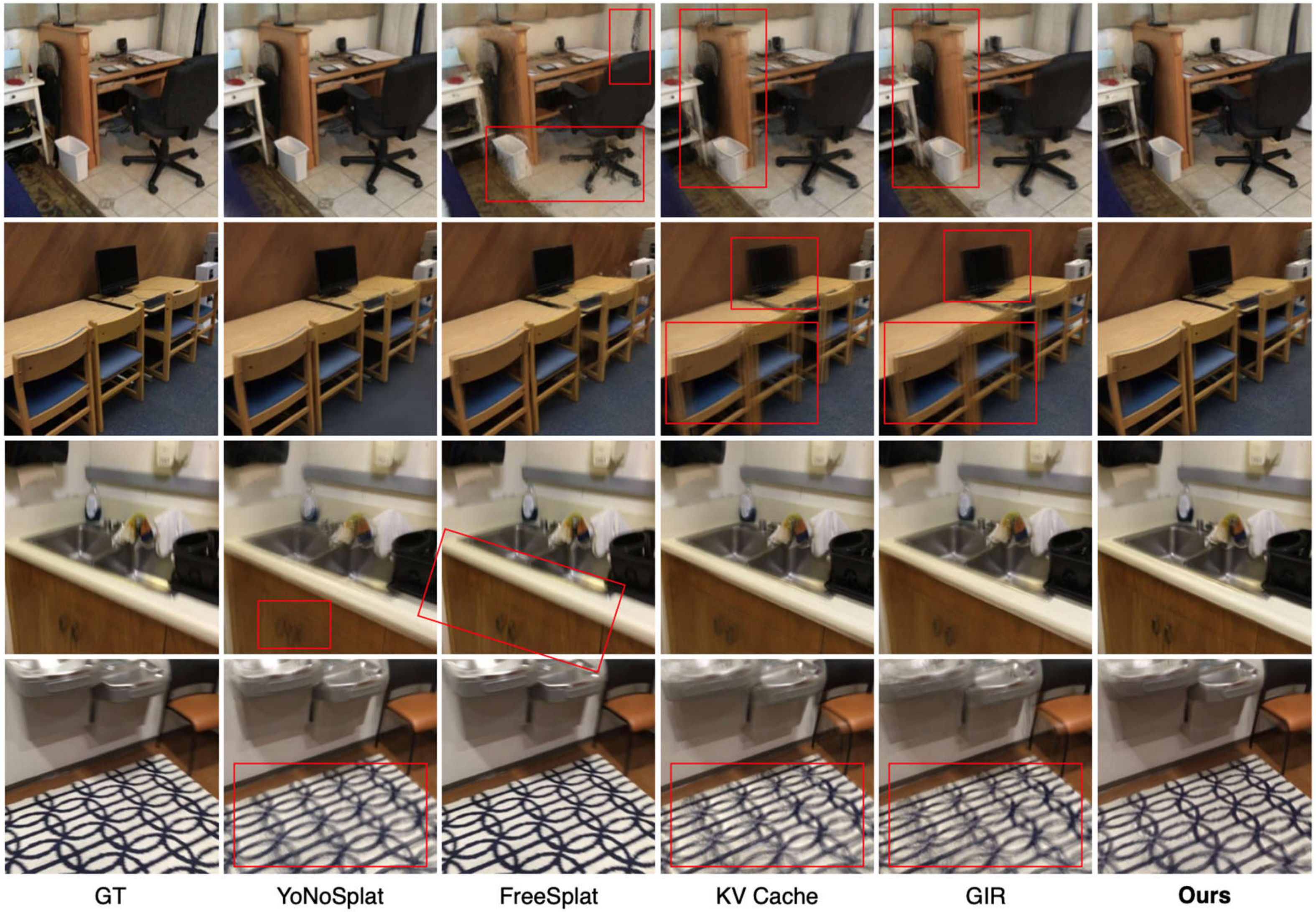


Method	Online	p	k	32v PSNR ↑	32v SSIM ↑	32v LPIPS ↓	64v PSNR ↑	64v SSIM ↑	64v LPIPS ↓	128v PSNR ↑	128v SSIM ↑	128v LPIPS ↓	256v PSNR ↑	256v SSIM ↑	256v LPIPS ↓
YoNoSplat	✗			22.308	0.736	0.180	22.253	0.732	0.183	21.827	0.720	0.194	20.749	0.677	0.226
KV Cache				21.705	0.703	0.202	21.302	0.690	0.223	20.794	0.655	0.247	19.829	0.606	0.292
GIR				21.752	0.703	0.203	21.382	0.692	0.222	20.933	0.660	0.245	19.990	0.608	0.291
Ours				**22.097**	**0.716**	**0.194**	**21.774**	**0.696**	**0.213**	**21.284**	**0.672**	**0.235**	**20.220**	**0.617**	**0.281**
YoNoSplat	✗		✓	22.575	0.748	0.177	22.504	0.747	0.179	22.053	0.735	0.189	20.963	0.692	0.221
S3PO-GS			✓	13.062	0.197	0.635	13.952	0.217	0.645	17.227	0.375	0.548	13.102	0.196	0.697
KV Cache			✓	21.932	0.717	0.198	21.544	0.697	0.218	20.993	0.670	0.242	20.087	0.622	0.286
GIR			✓	21.947	0.716	0.199	21.680	0.697	0.218	21.107	0.672	0.241	20.085	0.622	0.286
Ours			✓	**22.417**	**0.734**	**0.188**	**22.068**	**0.713**	**0.206**	**21.576**	**0.690**	**0.227**	**20.430**	**0.633**	**0.275**
YoNoSplat	✗	✓	✓	22.998	0.781	0.162	22.978	0.784	0.161	22.597	0.779	0.167	21.549	0.749	0.190
KV Cache		✓	✓	22.392	0.758	0.177	22.200	0.754	0.184	21.731	0.739	0.199	20.694	0.699	0.235
GIR		✓	✓	22.478	0.760	0.177	22.264	0.755	0.185	21.779	0.739	0.200	20.743	0.699	0.235
Ours		✓	✓	**23.084**	**0.780**	**0.164**	**23.098**	**0.782**	**0.167**	**22.853**	**0.777**	**0.176**	**22.003**	**0.751**	**0.202**

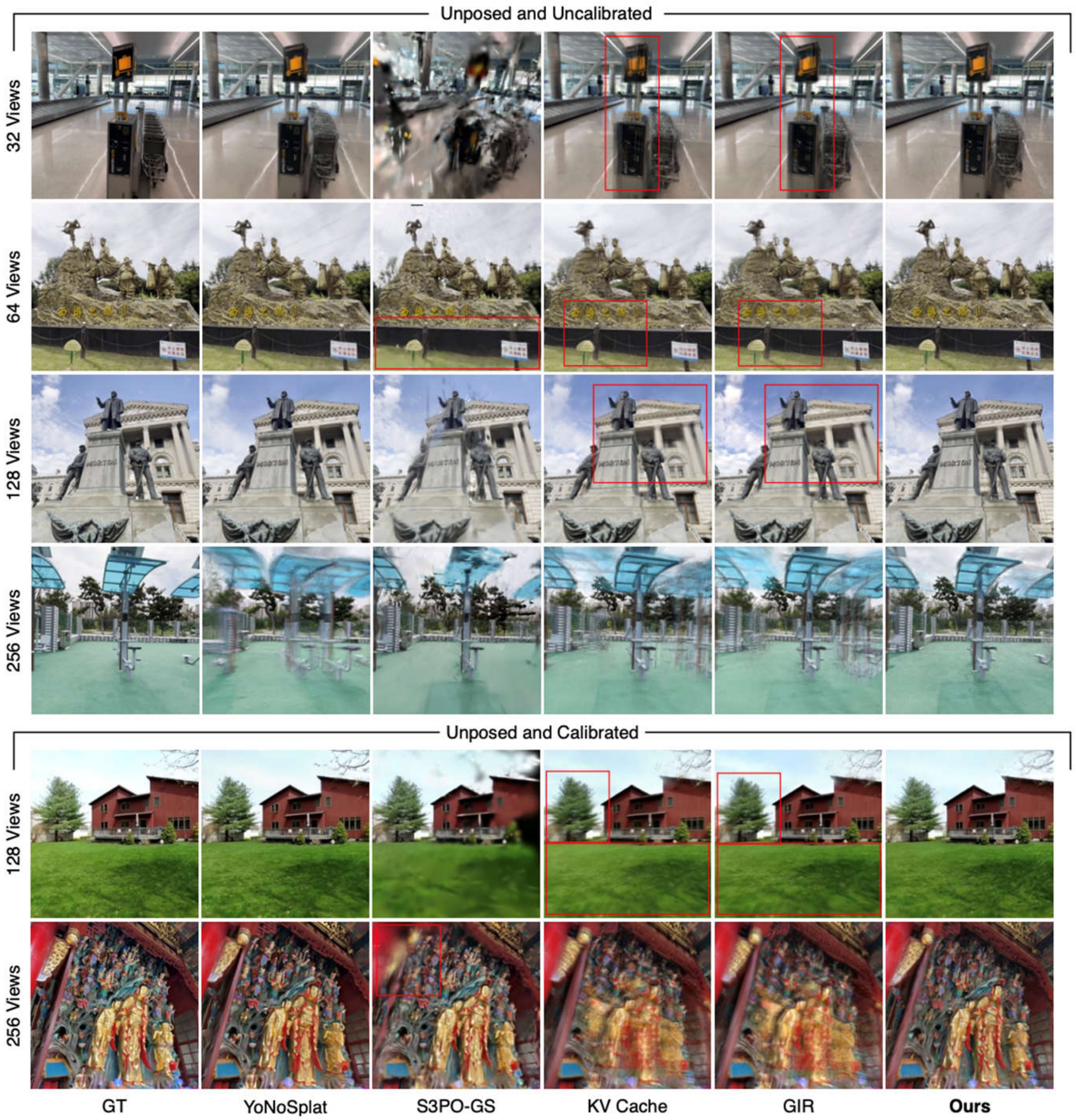
Method	Online	ACID						RealEstate10K						DL3DV					
		32v			64v			64v			128v			128v			256v		
		5° ↑	10° ↑	20° ↑	5° ↑	10° ↑	20° ↑	5° ↑	10° ↑	20° ↑	5° ↑	10° ↑	20° ↑	5° ↑	10° ↑	20° ↑	5° ↑	10° ↑	20° ↑
YoNoSplat	✗	0.493	0.670	0.787	0.475	0.661	0.783	0.701	0.846	0.921	0.685	0.834	0.911	0.741	0.870	0.935	0.738	0.869	0.935
CUT3R		0.163	0.379	0.587	0.105	0.312	0.528	0.282	0.547	0.754	0.140	0.411	0.666	0.106	0.398	0.670	0.005	0.024	0.244
TTT3R		0.193	0.409	0.614	0.159	0.376	0.588	0.338	0.591	0.776	0.270	0.539	0.748	0.524	0.754	0.877	0.263	0.579	0.785
StreamVGGT		0.028	0.153	0.382	0.018	0.124	0.348	0.021	0.164	0.428	0.014	0.133	0.393	0.012	0.146	0.427	OOM	OOM	OOM
WinT3R		0.042	0.177	0.381	0.016	0.094	0.252	0.186	0.465	0.700	0.077	0.294	0.562	0.100	0.372	0.648	0.011	0.117	0.389
Ours		**0.444**	**0.635**	**0.764**	**0.406**	**0.607**	**0.749**	**0.689**	**0.839**	**0.917**	**0.666**	**0.823**	**0.905**	**0.715**	**0.857**	**0.928**	**0.709**	**0.854**	**0.927**

Fully posed

Fully posed

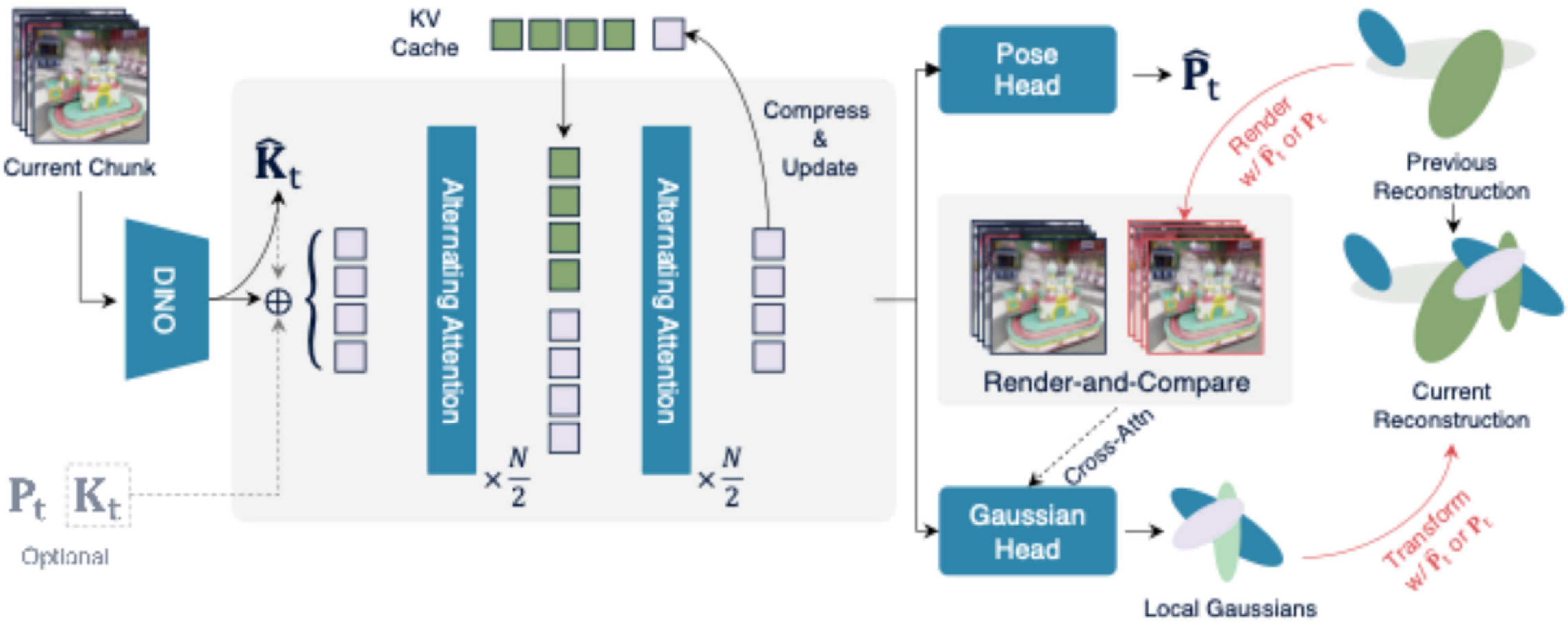


Unposed



Unposed and Uncalibrated
32 Views
64 Views
128 Views
256 Views
Unposed and Calibrated
128 Views
256 Views
GT
YoNoSplat
S3PO-GS
KV Cache
GIR
Ours

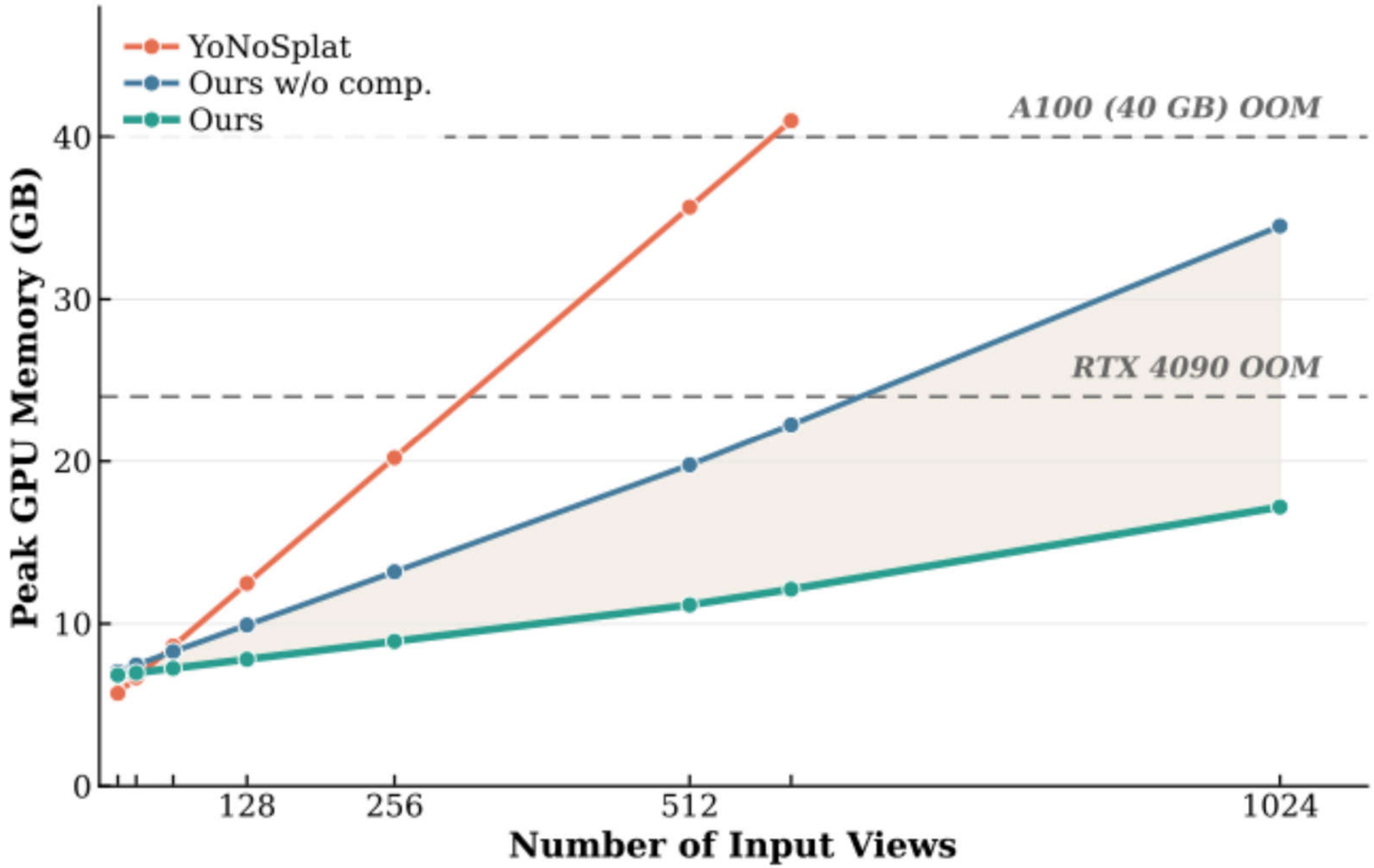
Scaling ReCoSplat to 100s of Images

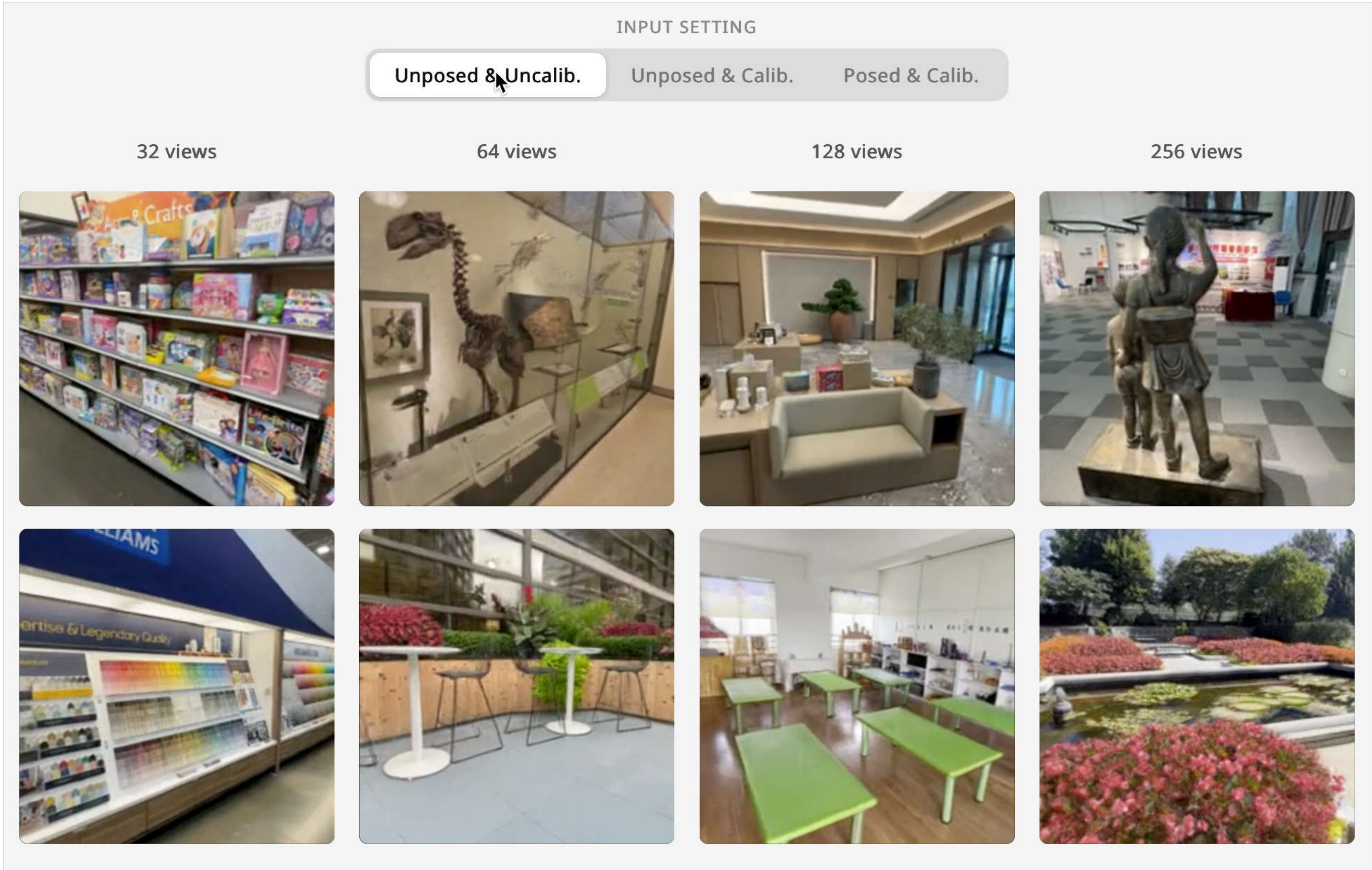


Scaling ReCoSplat to 100s of Images

- **Early Layer Truncation**: Because early global attention layers primarily extract local features rather than establishing multi-view correspondences, we completely discard the KV caches for the first 10 layers.
- **Selective Context Retention**: For the remaining 8 layers, we process frames in chunks. After an initial chunk establishes context, we retain historical tokens from only a single representative view. Specifically, the final frame of each subsequent chunk.
- **Trainable Register Token**: To ensure the network effectively utilizes this compressed cache, we introduce a trainable prompt register token to explicitly mark these retained views.

Scaling ReCoSplat to 100s of Images




INPUT SETTING
Unposed & Uncalib.
Unposed & Calib.
Posed & Calib.
32 views
64 views
128 views
256 views

Appendix: Alternate Conditioning Strategies

Method	Stage	p	k	PSNR ↑	SSIM ↑	LPIPS ↓
KV Cache	1		✓	21.346	0.680	0.224
KV Cache + Predicted Assembly Pose	1		✓	20.781	0.641	0.238
KV Cache + Mixed Assembly Pose	1		✓	21.316	0.681	0.223
KV Cache + Plücker Raymap	1		✓	21.166	0.676	0.228
Ours w/o Gaussian Features	1		✓	<u>21.455</u>	<u>0.685</u>	<u>0.223</u>
Ours	1		✓	**21.769**	**0.699**	**0.212**
KV Cache	1	✓	✓	21.866	0.729	0.196
KV Cache + Predicted Assembly Pose	1	✓	✓	20.836	0.656	0.229
KV Cache + Mixed Assembly Pose	1	✓	✓	21.915	0.736	0.192
KV Cache + Plücker Raymap	1	✓	✓	22.067	0.744	0.189
Ours w/o Gaussian Features	1	✓	✓	<u>22.548</u>	**0.759**	<u>0.181</u>
Ours	1	✓	✓	**22.610**	**0.759**	**0.178**